Foil priority from broadcast video: three measurement protocols, one deployed artefact, and thirty-nine closed avenues

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Abstract

We report what a retrospective classifier of foil priority achieves on a corpus of 34 374 broadcast clips from 21 tournaments, and we report it under three protocols that are kept apart on purpose. Under *Protocol A* — 18 development tournaments, leave-one-tournament-out, no abstention, $n = 3047$ two-light touches — a decision tree over 40 stable features reaches 69.68%, a small temporal network averaged over three runs reaches 72.56%, and the deployed blend of the two reaches 73.88%. That last figure belongs to one frozen artefact and sits about half a point above what the method returns on average; measured with a fourth run it returns 73.32–73.65%, so the defensible improvement over the previous state is about $+3.7\,pp$, not $+4.2\,pp$. Under *Protocol B* — the same models behind a conformal gate at 90% nominal coverage — accuracy when the model speaks is 80.5% pooled, but the guarantee is marginal and not per-venue: across the 18 development venues realised accuracy runs from 64.8% at the worst to 91.2% at the best, only 10 of them clear the target, and coverage falls to 51.4%. Under *Protocol C* — all 21 tournaments, 4 958 two-light pairs, 413 features, no gate — accuracy is 69.17% $[67.76, 70.63]$ at the 16 venues where the scoreboard overlay can be read and 64.24% $[62.81, 66.91]$ at the five where it cannot, a gap of $4.93\,pp$ $[2.14, 6.97]$, $p(\leq 0) = 0.001$. No single one of these numbers is "the accuracy". Alongside the positive results we publish the negative ones: thirty-nine avenues that were closed, thirty-eight of them built and measured against their own null control and one closed by argument and labelled as such, under an explicit noise floor of $0.8\,pp$ established by feeding the model a feature block that is empty by construction and watching it gain $+0.69\,pp$. One of the thirty-nine cleared that floor and a Bonferroni-corrected threshold and still did not earn a product change, and we report it at length because the three reasons are not statistical: a seed-pooling convention, a blend weight, and a downstream calibration stage that was already collecting the benefit. The binding constraint on the whole project is statistical power: on the held-out set as it stands, the power to detect a true $+1.6\,pp$ improvement is 5.9%; resampling whole tournaments instead, the same target reaches 80% power only at about 20 tournaments.

This document has the form of a research paper because that form makes the evidence auditable. It is an internal technical report. It has not been peer reviewed, it is not a certification of refereeing accuracy, and the labels it is trained and scored against are the archive's record of who was awarded the touch, not the annotation of a human referee.

1 Introduction

In foil, a touch is not awarded to whoever closes the electrical circuit first. It is awarded to whoever held priority in the phrase — the attack that was established, the parry that answered it, the line that was taken — and the scoring apparatus only reports that two lamps lit and roughly when [3]. The apparatus is a witness to contact, not a judge of precedence.

That distinction is not a technicality; it is the whole difficulty, and it is measurable. Figure 1 shows it on 194 two-light clips: the fencer who was awarded the point had lit *second* in 104 cases and first in only 55, with a median lead of −33 ms. The correlation between arriving first and taking the point is negative. A rule that simply predicts the sign of the inter-lamp delay scores 61.33% over 3 130 development pairs — above chance, because the delay is weakly informative about who was attacking, but pointing the wrong way often enough that it is worse than useless as a shortcut.

This report answers one question — *given a broadcast clip that ends in a double-lamp touch, which side was awarded it?* — and reports the answer three times, because there are three defensible ways to ask it and they give three different numbers. Section 2 describes the corpus and the acquisition failure that splits it. Section 3 describes the measurement chain. Section 4 defines the three protocols and the noise floor that decides when a difference is real. Section 5 gives the results. Section 6 covers what the same chain produces besides priority. Section 7 is the longest and the most important: thirty-nine things that were tried and did not work, with the measurements that closed them. Section 8 states the limits.

What this report does not claim. It does not claim to referee. The action categories of Section 6.1 are geometric candidates derived from pose and timing, never a referee's call. The stature and distance figures of Section 6.3 are estimates with an uncertainty, not laboratory measurements. There is no human-annotated reference set anywhere in this project, so no number here is validated against expert judgement.

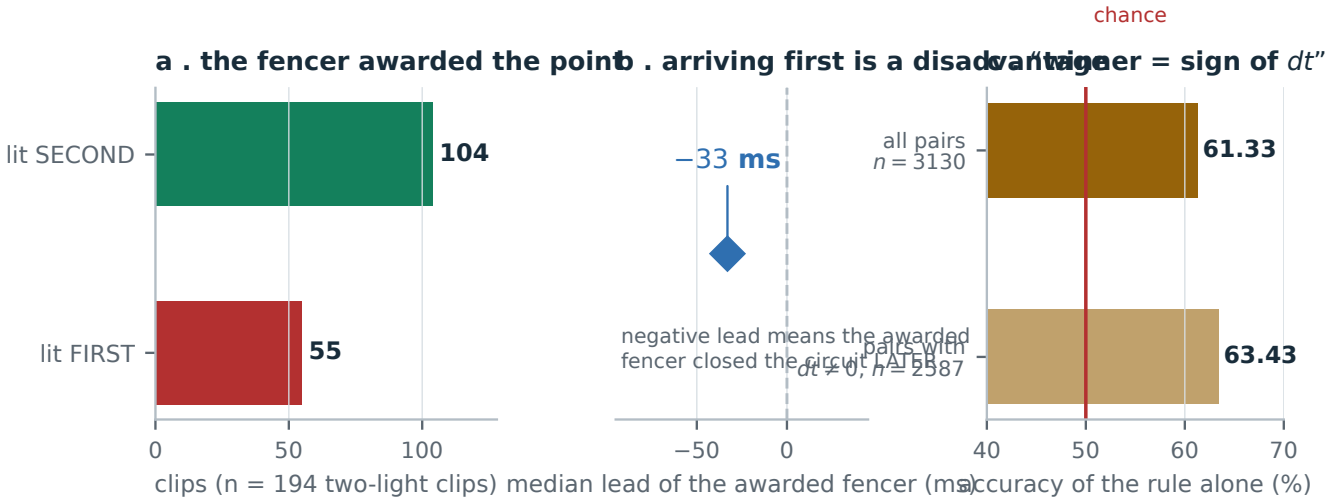


Figure 1. The apparatus does not referee the phrase. **(a)** On 194 two-light clips the fencer awarded the touch had lit second almost twice as often as first. **(b)** The median lead of the awarded fencer is −33 ms: negative, meaning the winner closed the circuit later. **(c)** Predicting the winner from the sign of the inter-lamp delay alone. Counts are the archive's record of who was awarded the touch, not a human referee's decision.

2 Dataset

2.1 Provenance and counts

The corpus is a manifest of 34 374 unique foil clips cut from broadcast footage of 21 tournaments, with identifiers running from `anaheimfoil2018` to `turinfoil2023`. Of those, 25 139 (73.1%) carry a usable left/right label. Figure 2 and Table 1 give the nested populations. The four numbers matter separately and are easy to conflate: the manifest is the inventory, the materialised set is what has measurements on disk, the two-light set is the priority target, and the development set is what the models in Section 5 were fitted on.

Two counting corrections are worth recording because both were silent. A directory of pose archives without an `index.csv` was discarded by the repository loader without an error, a warning or a count, even with a valid `index.jsonl` beside it; rebuilding the four missing index files took the materialised corpus from 15 452 to 23 011 clips and the fencer count from 354 to 401. And a single dictionary entry in any of twenty producer files changes the structural hash that each measurement artefact records, which invalidates all 23 011 at once; this has happened twice, and re-materialisation costs about twelve minutes across twelve processes. Neither of these is a modelling problem, and both changed the denominators of every number downstream.

2.2 Five venues where the scoreboard cannot be read

The single most consequential property of this corpus is that its acquisition quality is not uniform, and the non-uniformity is binary rather than graded.

The temporal anchor of first choice reads the lamp strip out of the federation's broadcast overlay, at a fixed normalised image band of $y\,0.910$–0.932. At 16 venues that band contains the lamp and the reader fires. At five — `kazanfoil2020`, `stpetersburgfoil2019`, `tokyoolympics2020`, `parisfoil2022`, `tokyofoil2022` — it does not, because those five broadcasts use four different overlay graphics: a capsule under the scoreboard (shared, identical to the millisecond, by the two Russian venues), a bar under a non-blue panel, a dark dot beside the digit, and a scoreboard cell that changes colour. Figure 3(a) plots where the lit lamp actually sits at each venue against the band the reader expects; Figure 3(b) plots how much of its region each overlay geometry fills when lit, against the two thresholds the detector applies. The capsule fills about 0.08 of its region against an event floor of 0.15, so the reader returns an empty list rather than a wrong answer. The remaining 16 venues fall back to the generic detector on between 0% and 11.5% of their clips: the failure really is a step, not a gradient.

Figures 4 and 5 show the three cases on real frames. In the readable case both coloured lamps are found inside the overlay and dated to the frame. In the fallback case the same clip is read by a generic

Table 1. Corpus census. The two-light population is the priority target; the one-light population is a different and harder task (Section 7.5).

Population	Clips
Manifest rows, unique foil clips	34 374
with a usable label	25 139
extraction succeeded	23 011
no lamp detected	2 038
pipeline error or pose failure	90
Measurements materialised	23 011
Unique clips in the repository	23 018
Two-light pairs, all 21 venues	4 958
16 venues, overlay reader works	4 217
5 venues, generic fallback	741
Two-light pairs, 18 development venues	3 130
One-light clips, development	11 460
Fencers with clips	401
Tournaments	21

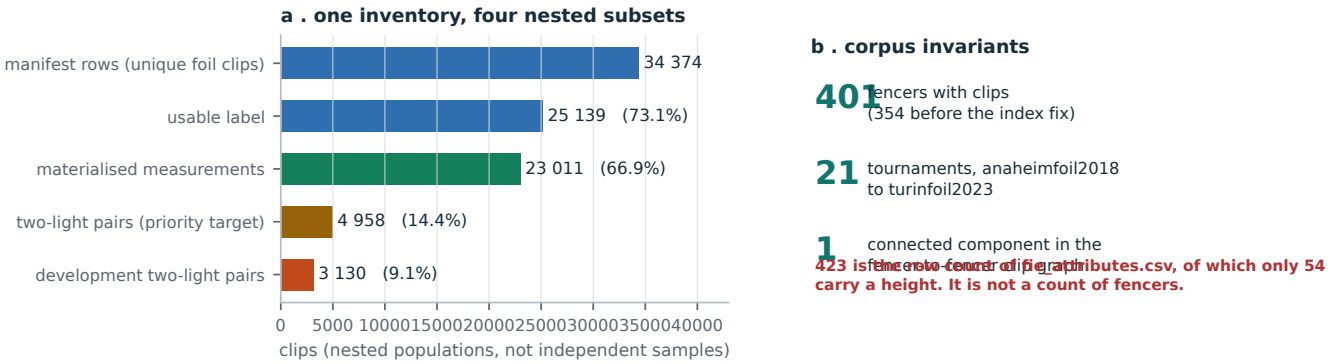


Figure 2. The corpus, as counted. **(a)** Four nested populations, not four independent samples: each is a subset of the one above it. **(b)** Invariants. The count 423, which appears in earlier internal documents as a number of fencers, is the row count of the federation attribute file, of which only 54 rows carry a height; the corpus contains 401 fencers with clips.

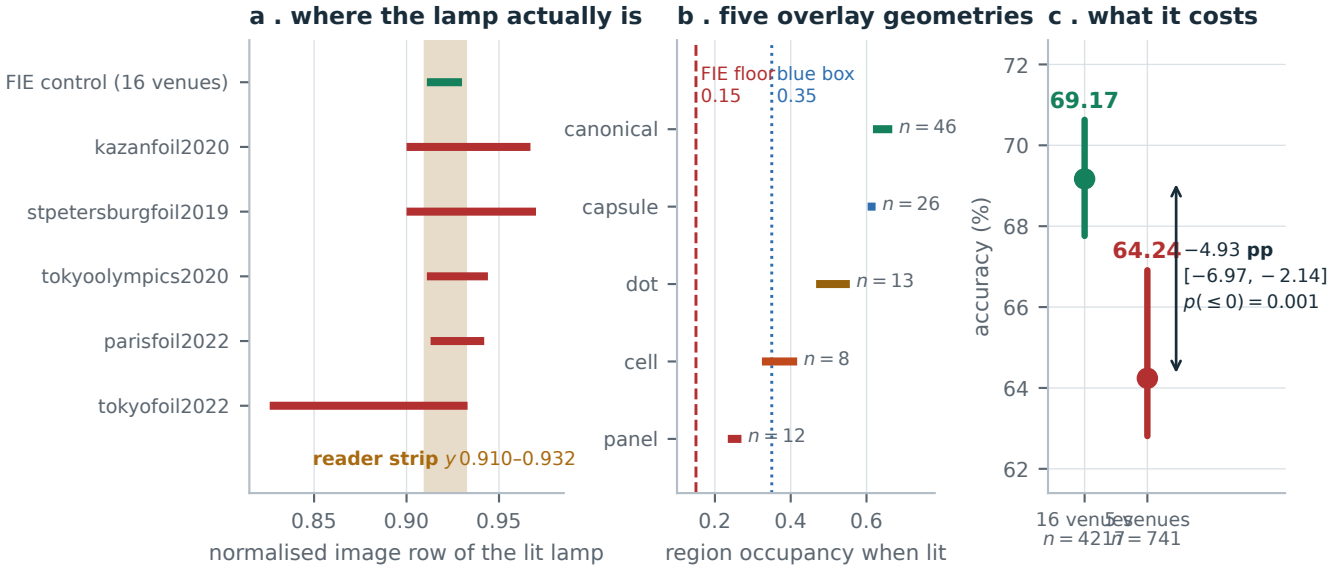


Figure 3. The acquisition failure that splits the corpus. **(a)** Measured normalised image row of the lit lamp, ten clips per venue, against the reader's fixed band (shaded). **(b)** Region occupancy when lit, by overlay geometry, against the event floor 0.15 and the blue-box threshold 0.35; n is the number of lamp events measured per geometry. **(c)** What it costs under Protocol C. This is a failure of data acquisition, not of the model.

colour detector over the whole frame, and the report for that clip carries the quality flags `shot_cuts` and `lights_from_generic_detector`. In the third case neither lamp produces an onset at all: both coloured channels are already on at frame 0 and switch off at 3.092 s, so there is no rising edge to date, and the only onset in the clip is a white lamp at 9.437 s.

2.3 The held-out split is confounded with the failure

The project's frozen tournament split assigns `parisfoil2022` and `tokyoolympics2020` to test and `stpetersburgfoil2019` to validation. All three are blind venues. Both held-out tournaments are therefore 100% generic-detector clips.

This has a consequence that no amount of modelling can remove: the accuracy drop on the held-out venues *cannot distinguish* between "the model does not generalise across venues" and "at those venues the touch instant is worse placed". Section 5.3 measures the second component directly and attributes about 1.4 of the 4.93 *pp* gap to the anchor. The remainder is not resolved, and we do not claim it is generalisation.

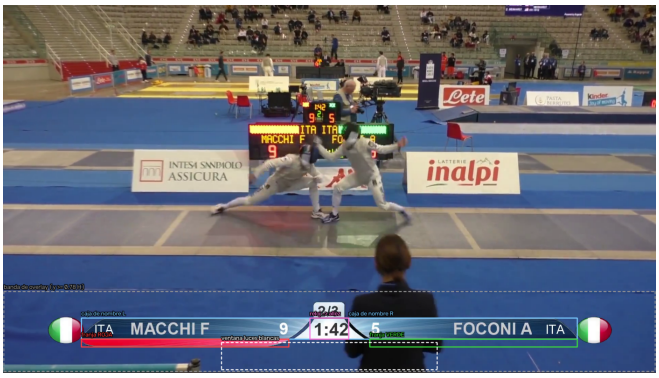
3 Method

3.1 The chain

Figure 6 lays out the pipeline. Each stage carries its own measured control, and the controls are reported here rather than assumed: they are the reason the numbers in Section 5 can be read at all.

3.2 Lamps

Lamp state is read per frame from four fixed regions — red at normalised $x = 0.332$, white-left at 0.377, white-right at 0.630, green at 0.675, symmetric about $x = 0.503$ — in rows 0.900–0.967 (Figure 7). Occupancy, the fraction of a region reading as lit, is the decision quantity. Over 292 clips from the venues where



(a) overlay readable



(b) generic fallback

Figure 4. Lamp detection on real frames. **(a)** `turinfoil2023`: the overlay is present in 100% of frames, the red left lamp is dated at 8.051 s and the green right lamp at 8.091 s, an inter-lamp delay of +40 ms, with region occupancies of 0.682 and 0.811. **(b)** `stpetersburgfoil2019`, one of the five blind venues: the same quantities have to come from a generic colour detector over the whole frame, and the clip carries shot-cut and generic-detector quality flags. Overlay annotations are drawn by the extraction tool; its labels *tirador izquierdo* and *tirador derecho* mean left and right fencer.



Figure 5. No datable onset. In this `turinfoil2023` clip (251 frames, 24.901 fps) both coloured channels are already lit at frame 0 and go off at 3.092 s, so neither produces a rising edge; the only onset is a white lamp at 9.437 s. 2 038 clips of the 25 139 attempted end in this state and are excluded, and the failure is concentrated by venue — 36.8% at `tokyofoil2022` against 2.5% at `tbilisifoil2023`.

the white channel was studied, the white lamp's occupancy has median 0.641, p90 0.655 and maximum 0.680, the same stability as the coloured lamps at about 0.62. What the white channel does *not* give is availability: it is observed at some point in 55 of those 292 clips (21.7%) but at the touch instant itself in only 14 (5.5%). The accept threshold applied to the white channel is not recorded in the source report, so none is quoted here.

The population split by lamp count is shown in Figure 8: 11 234 one-lamp clips against 3 203 two-lamp clips in the development network. Only the second population carries a priority question.

3.3 The temporal anchor

Everything downstream is measured relative to the instant of the touch, so the anchor is the most load-bearing single quantity in the chain. Four candidates were implemented and compared on 160 clips drawn ten per venue from the 16 readable venues, of which 153 have an independent reference instant (Table 2, Figure 9).

Table 2. Anchor candidates, $n = 153$ clips with a reference instant. Coverage is the fraction of clips on which the candidate fires at all. The bout clock's floor is structural: it changes once a second, so its best achievable median is 250 ms.

Anchor	Cov. (%)	Median (ms)	p90 (ms)	≤ 100 ms (%)
Per-region reader	94.8	0	0	100.0
Generic detector	98.7	0	0	92.7
Bout clock	90.2	360	2 052	18.8
Scoreboard jump	100.0	2 928	5 333	2.0

The per-region reader was adopted. Its worst case over those 153 clips is 100 ms and it is frame-exact in 144 of the 145 clips on which it fires. The scoreboard transition, which is the most obvious candidate because it is the most legible thing on screen, lags the touch by 2 928 ms at the median and is within tolerance on 2.0% of clips. The bout clock cannot be fixed by better reading.

Because 23 011 clips had already been extracted with the older anchor, the existing corpus was audited rather than assumed: 330 clips, 110 from each of the three blind venues whose video was already local, comparing the stored touch frame against the region reader. The result is 87.9% frame-exact (290 clips), 2.7% wrong (9 clips, deviating by a median of 3 301 ms and up to 7 169 ms, and all nine at a single venue) and 9.4% (31 clips) not verifiable either way. The stored window still contains the true instant in all but 3 of the 330.

3.4 The window

Features are computed over a fixed window of PRE_S = 4.0 s before the anchor and POST_S = 0.3 s after it, which is 129 frames at 30 fps (Figure 10(a)). The window is fixed rather than adapted; that choice is itself an ablation and is reported in Section 7.2.

The window is not a free parameter, because the an-

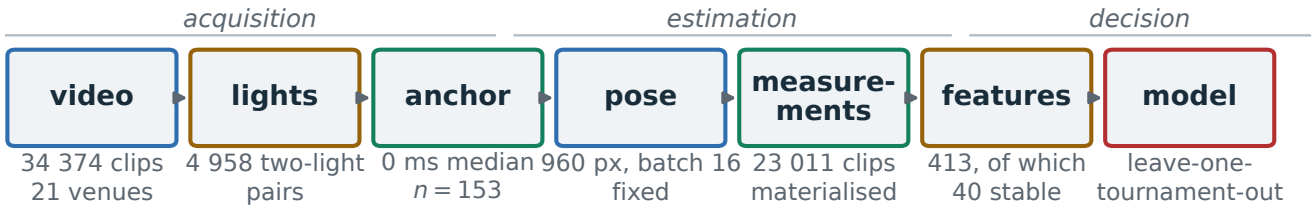


Figure 6. Video to decision. The quantity under each box is that stage's own measured control, reported in the subsection indicated in the text. A failure at any stage propagates downstream: the five blind venues fail at "lights" and the cost is paid by the model.

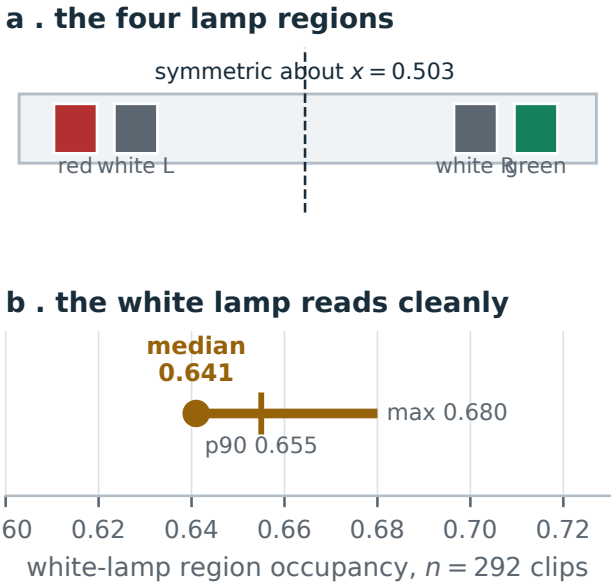


Figure 7. **(a)** The four fixed lamp regions, symmetric about the frame centre. **(b)** White-lamp occupancy over 292 clips. The lamp reads cleanly when it is there; the problem is that at the touch instant it is there in only 5.5% of clips.

chor error feeds straight into it. Figure 10(b) measures the cost directly: closing the feature window 0.667 s earlier than the anchor costs 4.2 *pp* on two-light clips (64.42 → 60.23) and 9.7 *pp* on one-light clips (63.32 → 53.67). This is also the sharpest available evidence that the models are using the phrase and not a broadcast artefact — an artefact would not care where the window ends.

Figures 11, 12 and 13 show what the window actually captures on real clips. The retained window is a small and varying fraction of the clip: median clip length 14.9 s against a median window of 4.28 s. The *allez* that starts the phrase is recoverable from the bout clock for fewer than half the clips, and 17% is irrecoverable, which is why the window is anchored on the touch at the end rather than on the start of the phrase.

3.5 Pose

Two fencers are tracked per clip with YOLO11m-pose [4] at 960 px, producing the 17 COCO keypoints [6] per fencer per frame, with left/right assignment done per frame and gaps filled by re-detection inside a region of interest.

The pose batch size is frozen at 16 and verified at runtime before every clip; it was 16 in 100% of rows across 434 extracted clips and in all 206 clips of a separate re-extraction audit, and the extraction loop has no dynamic batch reduction and no out-of-memory fallback, so it cannot silently change. At a fixed window the pose output is reproducible exactly: over 56 control clips with identical anchor and length the maximum pose difference between two extractions is 0, and the same holds for the bounding boxes, the pan estimate and the duplicate mask. Where the window grew, poses agree on 99.42% of visible joints.

It is worth being precise about what this does and does not establish, because the frozen batch has been treated internally as a hard constraint. What is measured is determinism *at* batch 16: two extractions of the same clip with the same window agree bit for bit. **No comparison between batch 16 and any other batch value exists anywhere in the measurement record.** The reason to keep 16 is therefore provenance rather than a measured sensitivity: the 23 011 materialised clips were all produced at 16, so changing it would make new extractions differ from the existing ones by an amount nobody has measured. That is a weaker justification than a measured displacement, and it also means the constraint has not been shown to be necessary. The experiment that would settle it is cheap — re-extract the same 56 control clips at a different batch and difference the pose arrays — and it has not been run.

Figure 14 and Figure 15 show the three regimes. In the clean case both skeletons are complete and every frame of the window comes from tracking. In the crossing case the fencers overlap, one skeleton loses five keypoints, and 176 frames of the window had to be recovered by region-of-interest re-detection with 24 frames missing outright. In the third case the confidence threshold rejects every keypoint of both fencers in the displayed frame even though clip-level coverage is high, and four frames are interpolated. These are not rare: pose quality is the second acquisition axis after lamp readability, and it is not uniform across venues either.

3.6 Measurements and features

From the pose series the pipeline materialises 68 measurements per fencer, and from those, 413 features per two-light pair. Forty of the 413 pass a stability criterion computed by leave-one-tournament-out over the venues

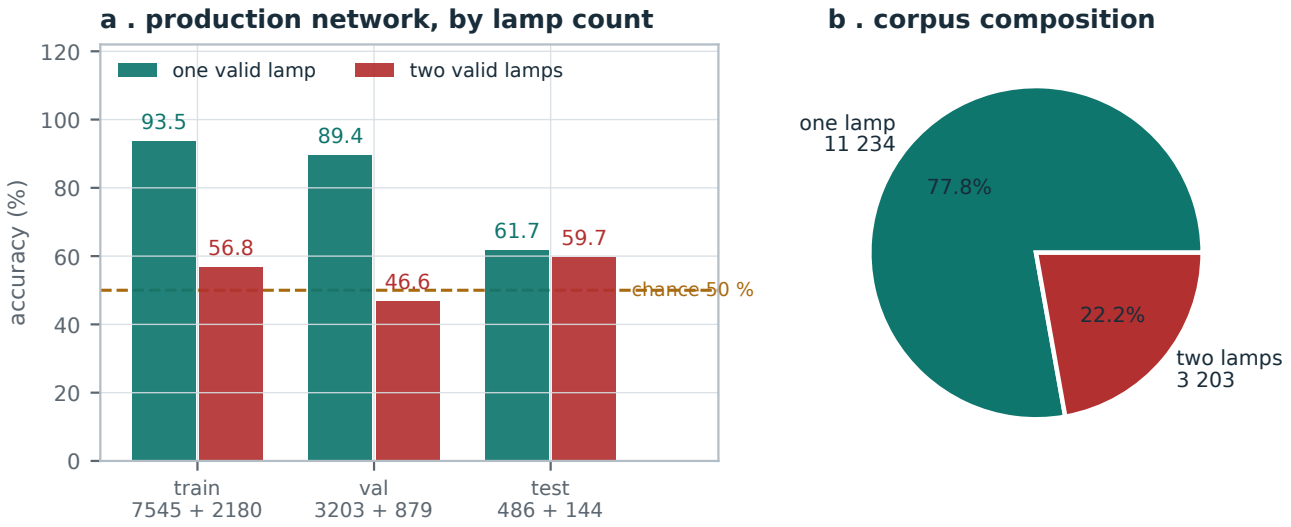


Figure 8. Why the two populations are not one problem. **(a)** Accuracy of the production network by lamp count and split. One-lamp touches are close to solved because the lamp itself carries the answer (93.5% on train, 89.4% on validation); on two-lamp touches the same network scores 46.6% on validation, which is below chance. **(b)** Corpus composition: 11 234 one-lamp against 3 203 two-lamp clips. Reused figure, re-rendered on white and in English. This is the older production network, not the models of Section 5; the comparison here is between populations, not between models.

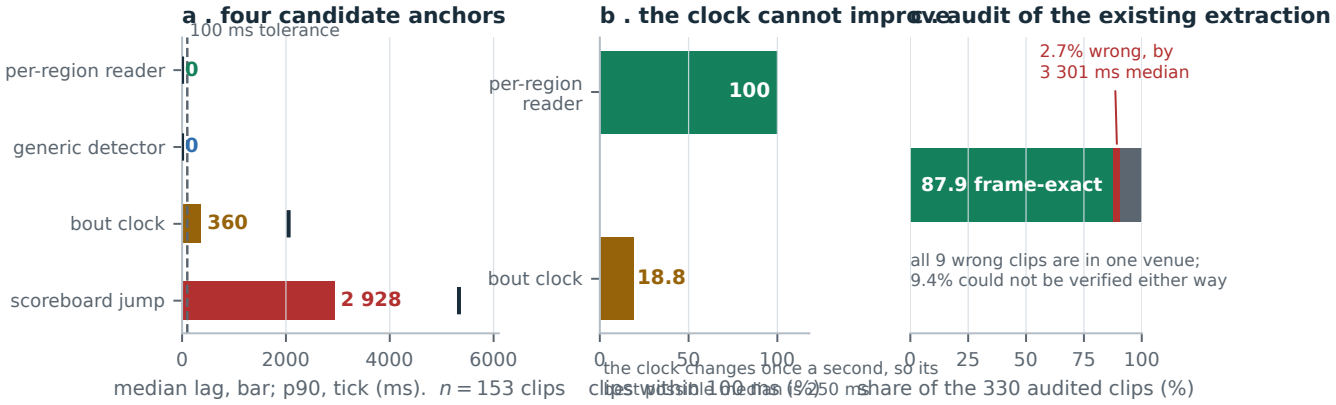


Figure 9. The temporal anchor. **(a)** Four candidates: bar is the median lag behind the touch, tick is p90. **(b)** The bout clock is limited by its own update rate, not by the reader. **(c)** Audit of the corpus that had already been extracted with the older anchor, 330 clips. Nine clips are wrong and all nine are at one venue.

with at least 30 pairs; that pruned set is what the deployed tree uses, and pruning to it is worth +1.22 *pp* (Section 7.4).

Two properties of this stage are worth stating because they bound everything else. Missing values are informative and must stay missing: filling the 0.81% empty cells of the 51-feature reference block with zeros before the tree costs exactly one point, 71.33 → 70.33%, which is above the noise floor. And the cost of the chain is dominated by decoding, not by inference: 53% of CPU time is video decode against 33% for the pose network (Figure 17), so the throughput lever is the codec path, not the model.

Figure 16 shows two independent quality checks on the output. The first is a sanity distribution: the on-guard distance, aggregated over 30 410 rows, has a median of 3.59 m, and 0.43% of rows are physically impossible — negative, or beyond the 8 m that the piste allows — which is the residue of the scale instability quantified in Section 6.4. The second is the state of the `gender` field over the whole 34 374-row manifest: it is unknown for 37.7% of rows, and seven of the 21 tournaments publish no sex field at all. That is the reason the sex-mean stature baseline of Section 6.3 was never usable everywhere, and it is a property of the source archive rather than of the pipeline.

4 Experimental protocol

4.1 Leave-one-tournament-out, and why grouping

All accuracies in this report are computed by holding out one whole tournament at a time (Figure 18(a)). The alternative — a random clip split — is not defensible here, and the reason is specific rather than generic: one venue is one camera, one broadcast overlay and one light-

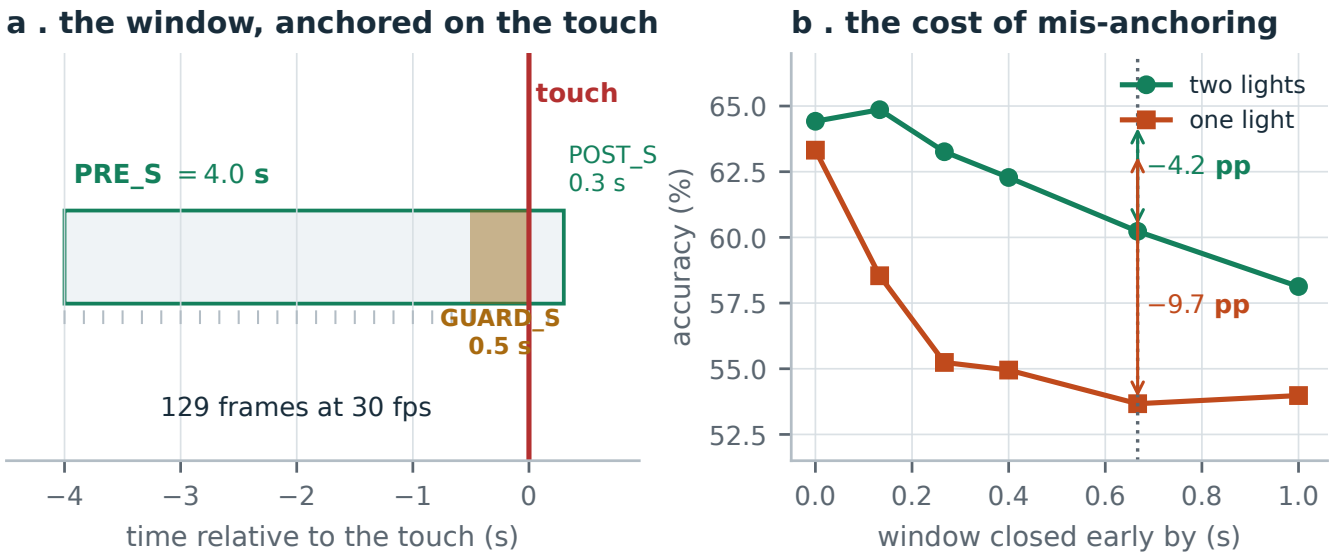


Figure 10. **(a)** The analysis window, anchored on the touch: 129 frames at 30 fps, with the 0.5 s guard segment used for on-guard geometry. **(b)** Measured cost of mis-anchoring, on the development two-light and one-light populations. The window is fixed by construction; this curve is the accuracy cost of moving it, not a confidence interval.

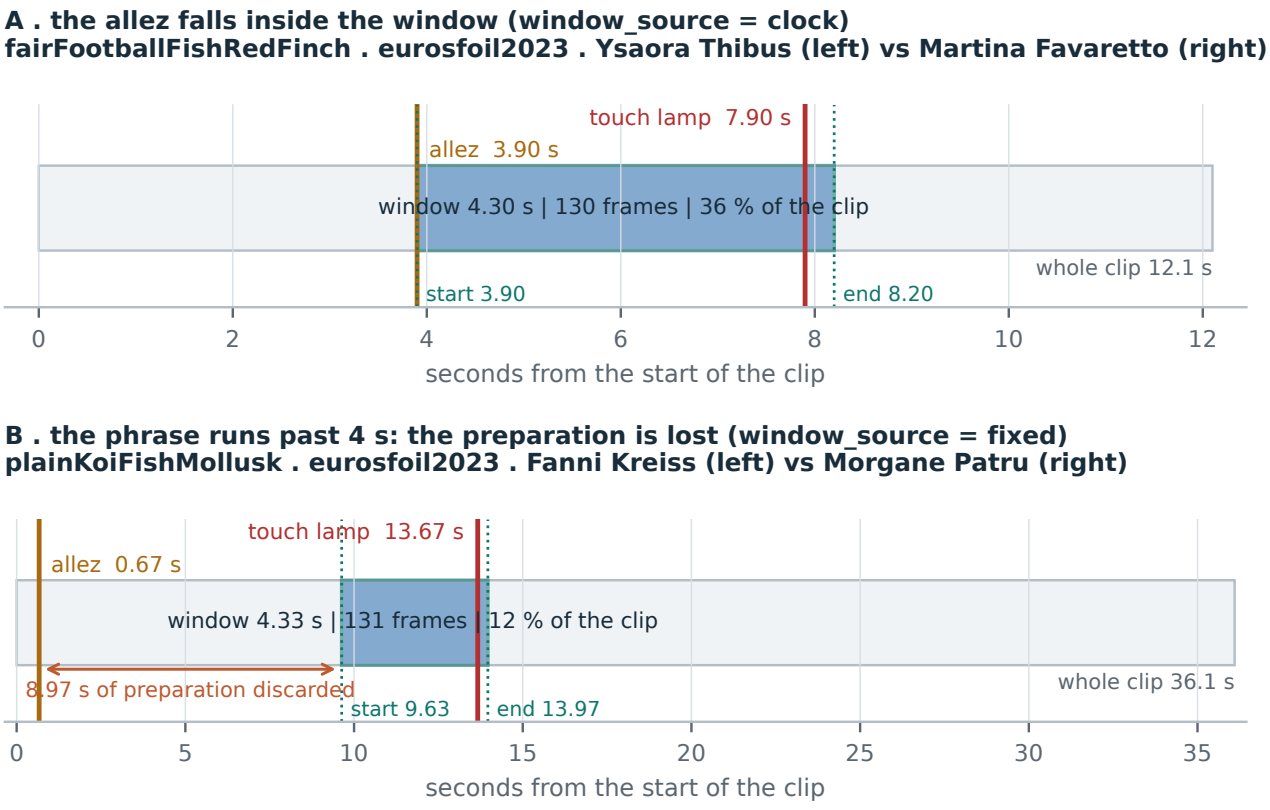


Figure 11. Two real clips end to end, showing why the window is anchored on the touch. In A the *allez* is recovered from the bout clock and falls inside the window. In B the phrase runs past 4 s and 8.97 s of preparation is discarded. Reused figure, re-rendered on white and in English.

ing rig, so clips from the same venue are not exchangeable with clips from another. A random split lets a model recognise the broadcast instead of the fencing.

The cost of grouping is visible: on the same features, a 5-fold grouped split gives 69.11% while leave-one-tournament-out gives 68.38% weighted and 68.76% as a mean over 18 venues, with a standard deviation of 5.37 *pp* and a range of 61.5–80.2%. The between-venue variance is 1.75× what independent sampling would produce. Figure 19 shows the per-venue spread and makes the confound of Section 2.3 plain: the two test tournaments sit at 61.3%, at the bottom of a validation range that reaches 94.1%.

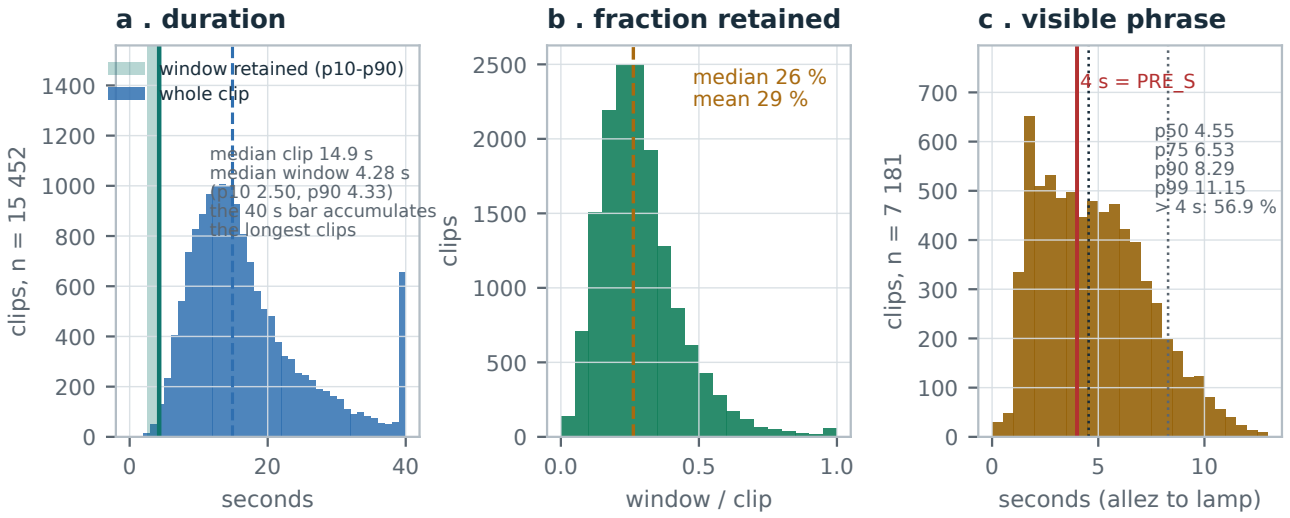


Figure 12. Distribution of clip length and of the retained window over the corpus. The window is a small fraction of most clips, and the fraction varies by a factor of three. Reused figure, re-rendered on white and in English.

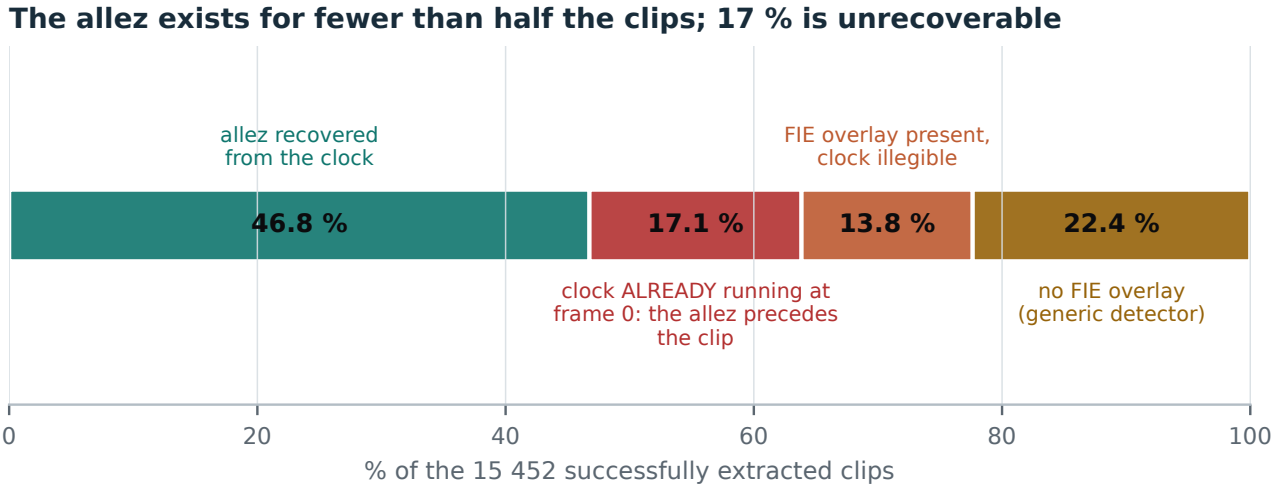


Figure 13. Why the phrase cannot be anchored on its start. The *allez* is available for fewer than half the corpus and 17% of it is irrecoverable; the bout clock is either absent, illegible, or already running at frame 0. Reused figure, re-rendered on white and in English.

4.2 Mirror and antisymmetry

Left and right must be treated identically, so instead of hoping the fit learns that, it is imposed. Each clip is also presented mirrored, and the prediction is the antisymmetric average

$$\hat{p}(x) = \tfrac{1}{2}\big(p(x) + 1 - p(m(x))\big), \qquad (1)$$

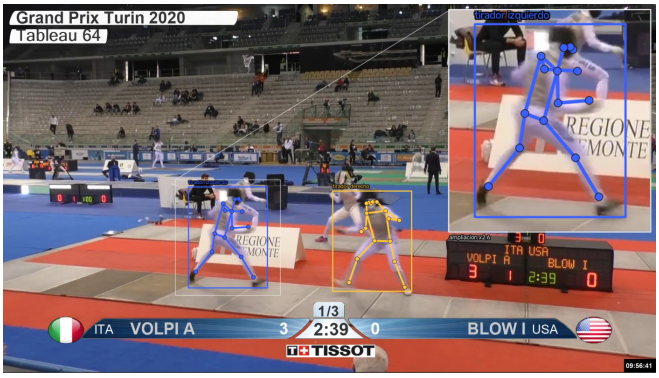
where m is the left/right mirror. The measured residual asymmetry of Equation 1 is 1.11×10^{-16} on both development and held-out data, i.e. exact to machine precision. This has a side effect worth noting: any feature that is symmetric under m has, by construction, no marginal signal, which is why the null block of Section 4.4 could be built at all.

Probabilities are calibrated with a symmetric Platt map [9]. Calibration improves the expected calibration error from 0.0219 to 0.0161 and leaves accuracy unchanged at 69.69%; isotonic calibration is slightly worse on both.

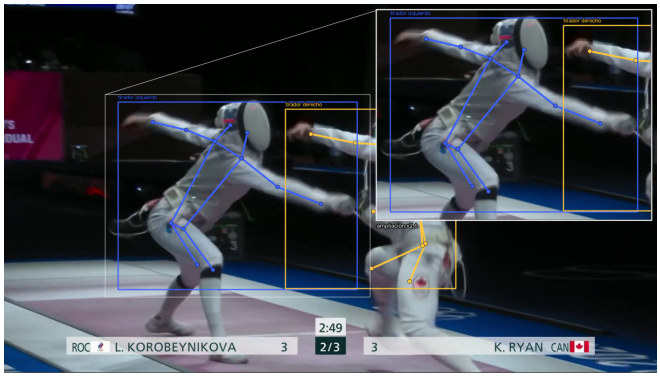
4.3 The conformal gate

For deployment the model is allowed to abstain. A split-conformal gate [1, 11] is calibrated with the calibration fold grouped by tournament, and the model answers only when the prediction set contains a single label.

The scope of the guarantee is the part that is easy to misread and is the subject of Section 5.2: it is *marginal*. It holds over the pooled calibration distribution, not conditionally on the venue. A per-venue conditional guarantee was not attempted, because with 18 venues it would be so loose as to say nothing.



(a) clean: 15/17 and 17/17 keypoints, all 158 window frames tracked



(b) crossing: 12 and 11 keypoints, 176 frames re-detected, 24 missing

Figure 14. Pose extraction on real frames, with the inset showing the enlarged fencer. **(a)** `turinfoil2020`, frame 157 of a 79-frame window: clip coverage 1.00 for both fencers and no interpolation. **(b)** `tokyoolympics2020`, frame 314 of a 215-frame window: the fencers cross, right-fencer coverage falls to 0.888, and most of the window is recovered by re-detection rather than tracking. Overlay labels are the extraction tool's; *tirador izquierdo* and *tirador derecho* mean left and right fencer.



Figure 15. A frame where the confidence threshold admits no keypoint at all for either fencer (0/17 and 0/17), although clip-level coverage is 0.958 and 1.00 and only 9 frames of the 215-frame window are missing. `tokyoolympics2020`, frame 299. A high clip-level coverage number does not mean every frame is usable.

4.4 The noise floor

A protocol needs a rule for when a difference is real, and here that rule is measured rather than assumed. Five null configurations were built and run through the full pipeline (Figure 20). The sharpest is a feature block that is null by construction: its features are symmetric under the mirror of Equation 1, so under the antisymmetric average they cannot carry marginal information, and the block's own declared discrimination is 0.5037. Adding it moves accuracy by $+0.69\,pp$. Circularly shifting the time axis of the three real blocks gives -0.22, -0.04, $+0.08$ and $+0.76\,pp$.

Mean $+0.25$, standard deviation 0.38, maximum $+0.76$. The floor is therefore taken as **0.8 pp**: no leave-one-tournament-out difference smaller than that is treated as an improvement anywhere in this report. That single rule is what turns Section 7 into a set of results instead of a set of excuses.

A sixth null, and the first that is not tabular. All five configurations above perturb tabular feature blocks, so the floor rested entirely on the tree side of the system. A sixth was measured on 2026-09-09 on the network side, and it perturbs nothing at all: *the same recipe, the same seed and the same interpreter, changing only the attention backend* from the memory-efficient kernel to the exact `math` one. It returns **+0.36 pp** $[-0.56, +1.27]$ and **327 of 3 047 clips decided differently**. Doing nothing to the model produces a delta larger than three of the five arms of a real hyperparameter sweep run on the same day.

And a sharper limit than the floor, expressed in decisions. The same null gives a resolution statement that percentage points hide. Changing only the attention backend altered **327 of 3 047 decisions**. The best real intervention measured on the network that night, label smoothing at three seeds, altered **571**. So **a no-op reproduces 57% of the decision churn of the intervention**. Read that way, the network's effective resolution is not the $0.8\,pp$ floor — which was calibrated on tabular blocks — but something closer to a point of accuracy, and anything below it is indistinguishable from recompiling. Churn is worth reporting alongside the delta for exactly this reason: two arms can share a point estimate and differ by a factor of five in how many decisions they disturbed, and the one that disturbed few is the one to believe.

Two further consequences follow. The $0.8\,pp$ floor is corroborated from the neural side and not only the tabular one, by a construction nobody would have thought to call a null. And any network comparison run across kernels, interpreters or machines inherits at least this much noise *before* any modelling difference is measured: the run-to-run spread of $1.35\,pp$ that this project had been quoting for the network falls to a measured **0.66 pp** once the backend is pinned, so half of what looked like inherent variance was this. The canonical interpreter and its number are recorded in the artefact notes, and the fix that removes the term is a four-line backend selection that costs no time and makes repeated runs bit-identical, $\max|p_1 - p_2| = 0.000$.

How the null is built matters as much as that it ex-

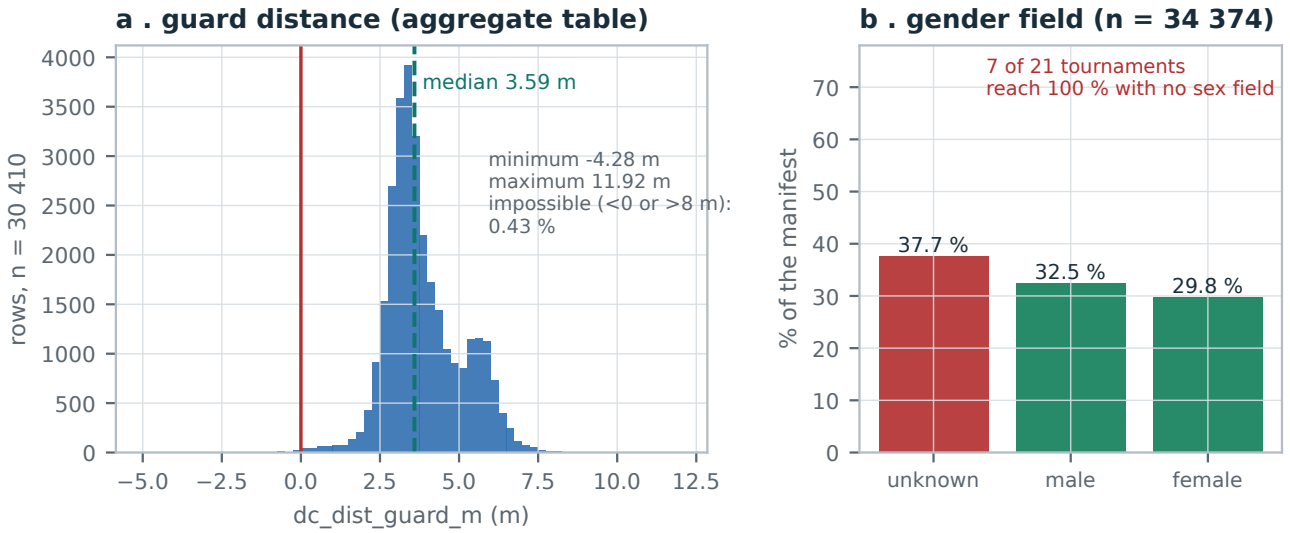


Figure 16. Two quality checks on the measurement output. **(a)** Distribution of the inferred on-guard distance over 30 410 rows, with the 0.43% of rows that fall outside what the piste physically allows. **(b)** Completeness of the `gender` field over the 34 374-row manifest. Reused figure, re-rendered on white and in English.

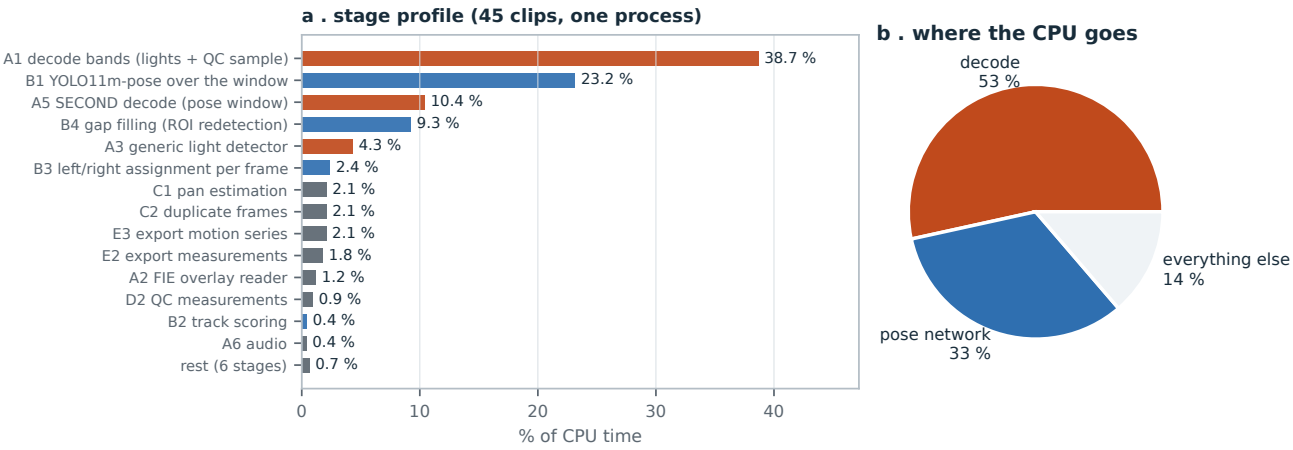


Figure 17. Where the CPU goes, profiled over 45 clips in a single process. Video decoding dominates. Reused figure, re-rendered on white and in English.

ists. Circular time shift keeps each trajectory smooth except at one seam and destroys only the alignment to the touch; the phase-randomised surrogate [10] keeps the power spectrum and the between-joint covariance and destroys the event structure. Flat frame shuffling was rejected for a measured reason: it introduces a discontinuity at every frame, which inflates exactly the high derivatives the features are built from. On the transverse-jerk peak, the median absolute difference is 1558.98 on real clips against 1926.15 on time-shuffled clips, a ratio of 1.24 — the shuffled null manufactures 24% more of the quantity it is supposed to destroy. That measurement is raw cache output with no n of its own and is an amplitude ratio on one feature family, not an accuracy effect; it is quoted here only as the basis for rejecting the construction.

4.5 The accepted nulls also fail, in two opposite directions

The argument of the previous subsection applies to the two nulls this report accepts, and they do not survive it. Both were audited column by column on 2026-09-09 against a 20-column block mixing level, count and derivative features, correlating each real column with its null on the antisymmetric channel that carries the side information.

The phase surrogate is sound: median $r = +0.044$, and **zero** of 19 columns above $r = 0.95$. The circular shift is not: median $r = +0.226$, but **three of 20** columns above 0.95, and the failure is structural rather than statistical.

Failure one: it preserves level features. A circular shift is a permutation of the window, so any statistic that does not depend on order is invariant *by construc-*

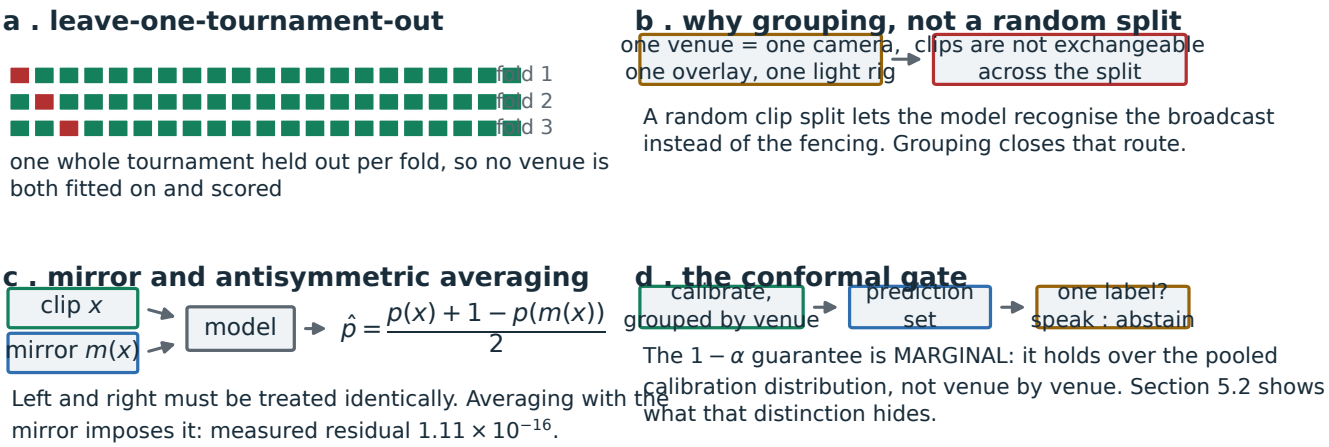


Figure 18. The four protocol choices, and the reason for each. **(a)** Leave-one-tournament-out. **(b)** Why the split is grouped by venue. **(c)** The mirror and the antisymmetric average. **(d)** The conformal gate, and the scope of its guarantee.

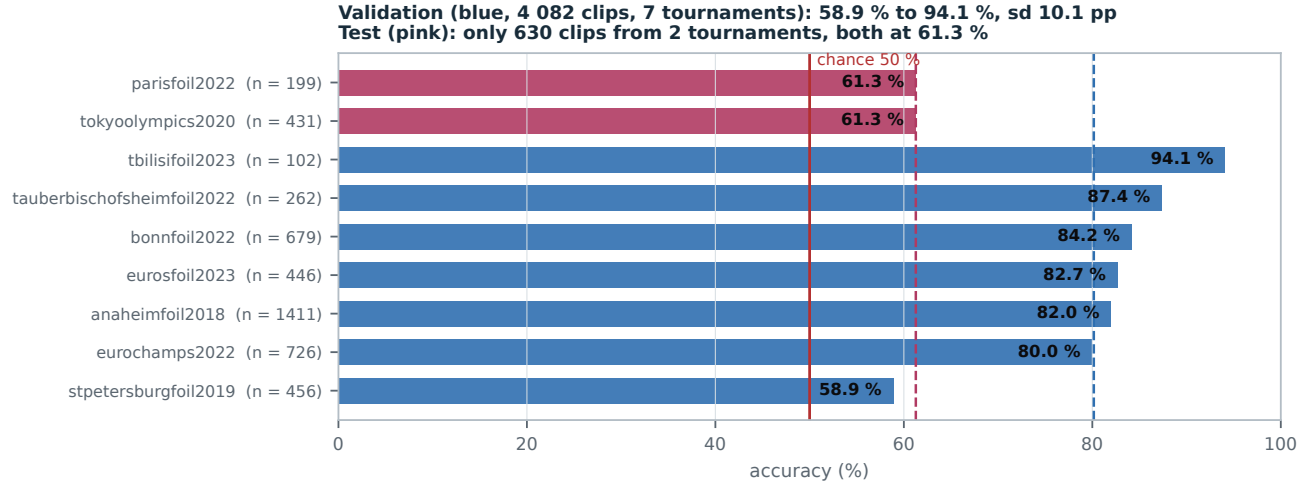


Figure 19. Accuracy per tournament. The spread across venues is much larger than the differences between models that Section 5 discusses, and the two test tournaments are both at the bottom of it. Reused figure, re-rendered on white and in English.

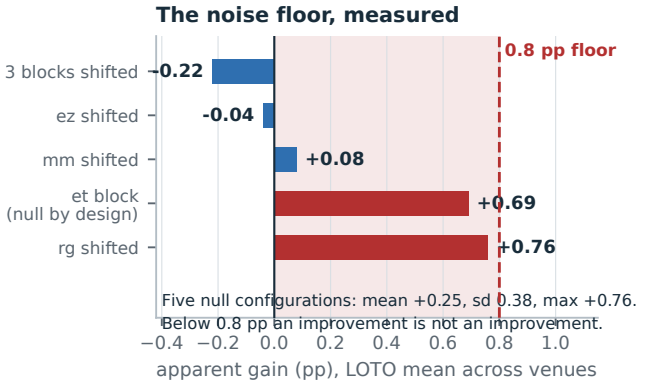


Figure 20. The noise floor, measured. Five configurations that cannot carry information, run through the whole pipeline. The largest gains $+0.76\,pp$ and the block that is empty by construction gains $+0.69\,pp$, which sets the floor at $0.8\,pp$.

tion. Measured: `d_ext_media` and `d_com_masa_media` both at $r = +1.000$, identical to the last digit, and `d_com_frac_avanza` at $r = +0.993$. That last one is the worst case available, because it is also the column with the highest univariate discrimination in the block, AUC $= 0.6615$ with all 18 venues above 0.50. For a block containing it, "does not beat its own null" means "does not beat a near-exact copy of itself", which is not a test.

Failure two: it inflates derivative features. The shift is applied to the positions and the features are derived afterwards, so the seam left by the roll becomes a large derivative. The null does not return a copy there; it returns something *bigger* than the real block: `d_com_a_min` at **7.79**× the real scale, `d_com_v_max` at 2.77×, `d_t83_amp` at 2.15×. Two failures of opposite sign in the same null, and **neither is visible in the aggregate metric**: the median scale ratio over the block is 1.00×.

One column behaves as the null intends, and its exception explains the rule. `d_com_v_media` comes out at $r = -0.369$ and 0.15× scale, properly destroyed, because the mean of a derivative telescopes to $(x_{\text{end}} - x_{\text{start}})/(n\,\Delta t)$ and therefore depends only on the two endpoints, which

a circular shift replaces entirely.

The null that works. For level features the correct control is a *random side-sign* null: it destroys the side information while preserving marginals, inter-column covariance and per-clip magnitude, and its correlation with the real block is −0.009. Run independently on a 32-column block it bought +0.82 *pp*, reproducing this report's own 0.8 *pp* floor from scratch by a different route — which is the strongest available evidence that the floor of Section 4.4 is the right number even though one of the nulls used to establish it is defective.

Failure three: it is inert on a pooled representation. A third mode appeared when the same null was applied to a learned embedding rather than to trajectory columns. Mean-pooling 1 568 tokens *already* destroys temporal order, so circularly reordering them before pooling changes almost nothing: the shifted embedding has cosine **0.9980** with the real one. Used over pooled representations this null tests nothing, and a block that "fails to beat" it has not been tested. The three modes have nothing in common except the name of the null, which is the point: **choose the null from what the representation actually encodes**, not from what the perturbation is called. A permutation of the time axis is a strong control for an event-locked feature, a near-identity for a window mean, an amplifier for a derivative, and a no-op for a pooled embedding.

Auditing a null needs three numbers, not two. Correlation answers "does it preserve?" and the scale ratio answers "does it inflate?". A third is required: **the sign of the null's own delta against the reference**. The floor of Section 4.4 was calibrated with a null that *helps*, +0.69 *pp*: inert columns buy a little. The side-sign null on an antisymmetric block *harms*, −1.03 *pp*, because shuffling the sign of a column that carries side information is not inert. The consequence is a scale error waiting to be made, and it was made once here before being caught: the quantity (real−null) is inflated by how damaging the control is and not only by how informative the block is, so **it does not live on the same scale as the floor**. The floor applies to the real block's delta against the reference, never to its excess over a harmful null. Every avenue in Table 5 was closed against the +0.69-calibrated floor, so any re-examination using the side-sign null must **recalibrate the floor with that null** rather than reuse 0.8 *pp*.

The recalibration, and the part of it that failed. The floor of the excess statistic was then measured at the centre-of-mass shape (24 columns, 12 of them antisymmetric) by applying the side-sign null to a block made inert by permuting rows within each venue, one permutation shared across columns. Over three permutation draws crossed with three sign draws the excess of inert over inert-plus-sign came out at −0.39, −0.43 and +0.03: **mean** −0.26 *pp*, **range** −0.43 **to** +0.03, **largest magnitude** 0.43 *pp*. Centre of mass excess is +1.42 *pp* with a per-draw range of +0.85 to +1.97, so **it clears the recalibrated floor**: on its own scale, that block's side information is real. The verdict is unchanged, because it is decided on the absolute delta (+0.39 *pp*, inside the 0.8 *pp* floor, interval crossing zero); what the recalibration adds is that the "signal is real" half of the redundancy finding is not an artefact.

The construction's mounting check failed, and the tool is therefore narrower than intended. The inert block scored −0.87 *pp* against the reference where this report's own inert block scores +0.69 *pp*: opposite sign. The cause is that permuting rows within a venue destroys the link to the label *and also* the row's internal coherence, so the model sees one clip's centre of mass beside another clip's 51 features. That is not inert, it is *contradictory*, and a contradictory block damages rather than helps. What survives is narrower and still useful: the excess is a difference between two versions of the same block, and the row incoherence is present equally in both, so it cancels. Hence 0.43 *pp* is the floor of the *excess* statistic and **not** a reproduction of this report's floor. *Consequence: the sign of the null's own delta is a valid threshold for the excess and nothing else. A permutation-inert block cannot be used to recalibrate absolute deltas; doing that would need the original inert construction rebuilt, which was not done.*

Two further limits. The figure is **specific to the block's shape**: the damage a side-sign null does scales with how many antisymmetric columns it corrupts, so a three-column family would have a smaller floor and there is one number per shape, not one number. And three draws give a range, not an interval. Note finally that no verdict in Table 5 depends on any of this: the scale correction above puts the floor on the effect column, and every row is decided there.

What this costs the report, which is less than it first appears. The t.83 block is well nulled by both constructions (median $r \approx 0.07$ on its columns, which are near-orthogonal at 0.07–0.14), so its closure at −1.12 *pp* stands on both criteria.

The centre-of-mass and pull blocks are the ones whose null was defective, and the first instinct is to re-open them. That instinct is wrong, and the convention stated at the head of Section 7 is why: an avenue closes when it fails *either* test. Centre of mass measured +0.39 *pp* [−0.87, +1.69] and pull −0.92 *pp*; **neither clears the** 0.8 *pp* **floor**, and the floor test does not involve a null at all. Both closures therefore stand on evidence the defect cannot touch.

What the defect cost is the *second*, corroborating line of evidence: Table 5 presents "worse than its own null" as though it were an independent confirmation, and for these two blocks it was not one.

The re-measurement, and what it changed. The side-sign null was then run on all four blocks with three independent draws each. Excess of the real block over its null: t.83 −0.77, pull −0.43, all three together +0.37 — all three closed against the healthy control too, so **those closures never depended on the defective**

one. Centre of mass came out at **+1.42 pp** of excess, which across the three draws ranges +0.85 to +1.97: the sign is stable, the magnitude is not, so it is quoted as *between one and two points*.

That single number changes the character of the verdict without changing the verdict. Two things are true together. The block **does carry genuine side information**: it beats, by one to two points, a version of itself with identical per-clip magnitude and identical inter-column covariance whose side is destroyed, and the circular null could not have seen this because it returned `d_com_frac_avanza` at $r = +0.993$. And the block **still does not improve the model**, $+0.39\,pp$ with the interval crossing zero, because that information is already inside by other columns: $r = 0.877$ with `d_cs_v_mean_signed`, 0.841 with the `fw_` family and 0.802 with `cs_v_peak_fwd_ms`.

So the ledger row moves from "closed because its null beats it" to **"closed by redundancy: the signal is real and the model already had it"**. That is a different and more useful row, and only the correct null could tell the two apart. It is also the clean illustration of the general lesson: **a null control has a type, a feature block has a type, and a mismatch produces a confident number that means nothing** — and it means nothing in *both* directions. Here the mismatch was hiding a real signal, not manufacturing one.

4.6 The three protocols

Table 3 states the three protocols side by side. They differ in population, in whether abstention is allowed, and in feature set. They answer three different questions and their numbers are not comparable with one another. Nowhere in this report is a figure from one protocol subtracted from a figure of another.

5 Results

5.1 Protocol A: the model, with no abstention

Table 4 and Figures 21(A) and 22 give the result. The deployed tree over the 40 stable features reaches 69.68% weighted; the temporal network, averaged over three runs with seeds fixed in advance, reaches 72.56%; the blend of the two at a fixed weight of 0.65 reaches 73.88%, an improvement of $+4.195\,pp$ over the tree with a bootstrap interval of $[+2.69, +5.75]$, positive in 100% of resamples and winning at 16 of 18 venues.

The artefact is not the method. The 73.88% belongs to one frozen artefact. It is not selection or overfitting — the three seeds were fixed before the runs — but the blend has run-to-run variation, and this particular trio landed on a fortunate point of the weight sweep. Measured with a fourth run scored in full (a run which scores 70.33% on its own, inside the normal range 69.74–71.09%), the blend gives 73.32% at the same weight and 73.65% at its own best weight. The four triples that can be formed give 73.88, 73.48, 72.99 and 72.83%, mean 73.29%.

So the honest claim is a band: **the method returns** 73.3–73.65%, and the improvement over the previous state is **about** $+3.7\,pp$. Quoting $+4.2\,pp$ subtracts one artefact from another and attributes the difference to the method. Figure 22 draws both arrows so the distinction is visible. For scale, $+3.7\,pp$ is about 4.6× the noise floor of Section 4.4, so the improvement is real; it is the third decimal place that is not.

5.2 Protocol B: with the conformal gate

Behind the gate at 90% nominal coverage, measured on 4 346 touches from the seven validation venues that neither model had seen, accuracy when the model speaks rises from 87.81% to 95.97% while coverage falls from 99.3% to 69.6%. The share of all touches on which the system returns a confident and wrong verdict falls from 12.10% to 2.80% (Figure 23(c)). On the two-light subset specifically, the pooled accuracy when speaking is 80.5% at 51.4% coverage.

The guarantee is marginal, and 80.5% is not a floor. Pooled, the gate delivers what it promises almost exactly: at nominal failure rates of 20, 15, 10 and 5% the realised rates are 20.04, 14.98, 10.04 and 4.97% (Figure 23(a)). Per venue it does not. At $\alpha = 10\%$ the realised failure rate runs from 4.4% at the best venue to 18.3% at the worst, and 10 of 18 venues clear the target. In accuracy terms the same fact reads: 64.8% at the worst venue, 91.2% at the best. The two statements are the same measurement expressed on different denominators: the failure rate is over *all* touches while the accuracy is over *covered* touches, so at roughly 52% coverage 18.3% unit failure is $18.3/52 \approx 35\%$ error among the answered ones, hence 64.8%.

Note the population change inside this subsection. The pooled figures above come from the 4 346 touches of the seven held-out validation venues; the per-venue breakdown in this paragraph necessarily does not, since it counts eighteen venues. It is the leave-one-tournament-out gate over the 18 development venues, a different population and a different split. The two are quoted together because they describe the same defect from two angles, but no figure from one should be subtracted from the other. A caveat on provenance: the per-venue numbers trace to a cached gate artefact over 3 118 pairs whose Platt slope is 0.878, which is the tree-only head rather than the deployed blend; recomputing it directly returns 64.79, 90.91 and 80.24% with 9 of 18 clearing under a strict inequality. The blend's own gate is less pessimistic, with a worst venue of 73.68% and 8 of 18 clearing. The conclusion of the paragraph is unaffected and in fact holds for both gates, but the specific worst-venue figure belongs to the tree.

The 80.5% is a mean across venues. It is not a lower bound for any venue, and it must not be quoted as one. A product deployed at one specific competition is exposed to that venue's number, not to the mean.

Table 3. The three protocols. Every accuracy in Section 5 is labelled with the protocol that produced it. The three are not comparable with one another: they differ in population, in feature set and in whether the model is allowed to abstain.

	Protocol A	**Protocol B**	**Protocol C**
Question	how good is the model	how good is it when it speaks	what the acquisition failure costs
Venues	18 development	18 development	all 21
Population	3 047 two-light touches	4 346 touches, 7 validation venues	4 958 two-light pairs
Split	leave-one-tournament-out	leave-one-tournament-out	leave-one-tournament-out
Features	40 stable, plus the temporal net	as A	413
Model	tree, net, and their blend	as A, behind the gate	gradient-boosted trees
Abstention	none	conformal, $1-\alpha = 90\%$	none
Mirror and antisymmetry	yes	yes	yes
Intervals	tournament bootstrap	—	tournament bootstrap, 4 000 reps
Headline	73.88% artefact, 73.3–73.65% method	80.5% pooled, at 51.4% coverage	69.17% vs 64.24%, gap 4.93 *pp*

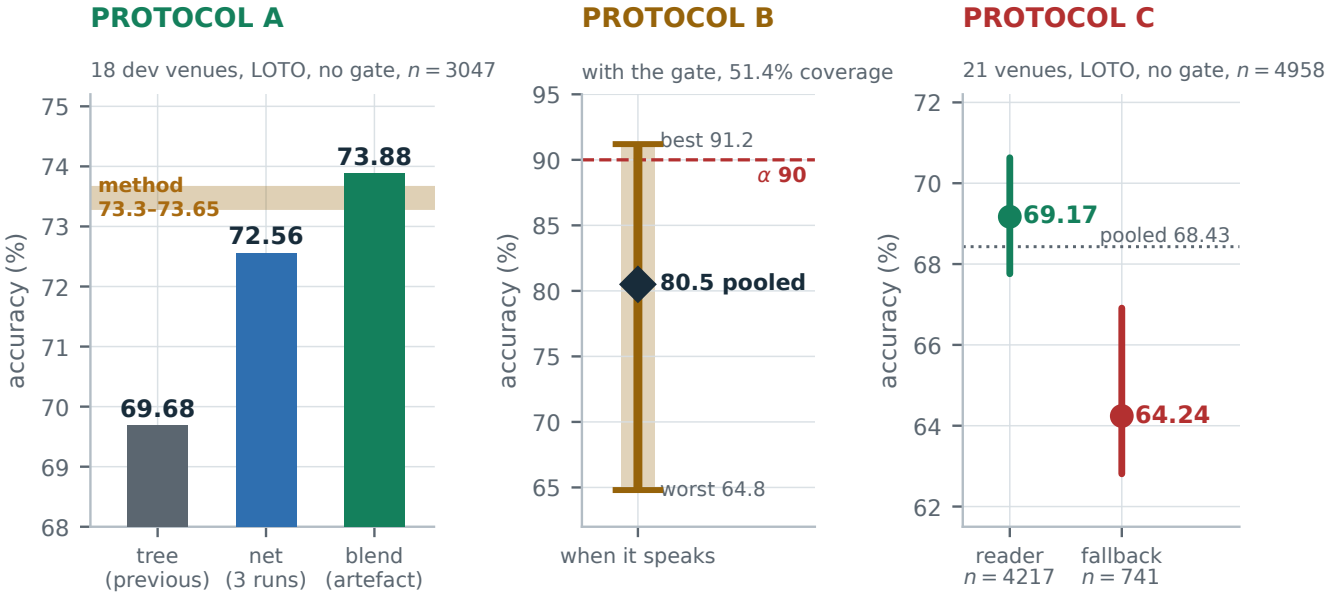


Figure 21. The three protocols. Each panel has its own population, its own n and its own vertical scale, and each is labelled with the protocol that produced it. In panel B the bar is the range of realised accuracy across the 18 venues, of which 10 clear the target: the diamond at 80.5% is the mean across venues and is not a lower bound for any of them. The panels answer three different questions; their numbers must not be compared with one another, and none of them alone is "the real accuracy".

Table 4. Protocol A. 18 development venues, leave-one-tournament-out, no gate, $n = 3047$ two-light touches. "Weighted" pools the touches; "mean" averages the venues.

Model	Weighted (%)	Mean (%)	Worst venue (%)
Tree, 40 stable features	69.68	69.89	62.41
Temporal net, 3 runs	72.56	72.28	66.18
Blend, $w = 0.65$	73.88	73.61	63.16
Blend, 4 runs, $w = 0.65$	73.32	—	—
Blend, 4 runs, $w = 0.76$	73.65	—	—
Mean of the four triples	73.29	—	—

The abstention is a feature, not a defect — a system that says "I do not know" on a third of touches and is right 96% of the time on the rest is more useful than one that always answers and is right 88% of the time. But the coverage cost has to be quoted with the accuracy gain, and it decomposes: of the 29.7 *pp* of lost coverage, 9.7 is conformal abstention on two-light touches, 5.8 is clips where an observed white lamp settles the question by other means, and 14.3 is clips where the white channel is unavailable at all — five whole venues, 22.4% of the corpus. That last block is the acquisition failure of Section 2.2 appearing again, in a different currency.

Figure 24 shows the trade as a curve rather than at one operating point.

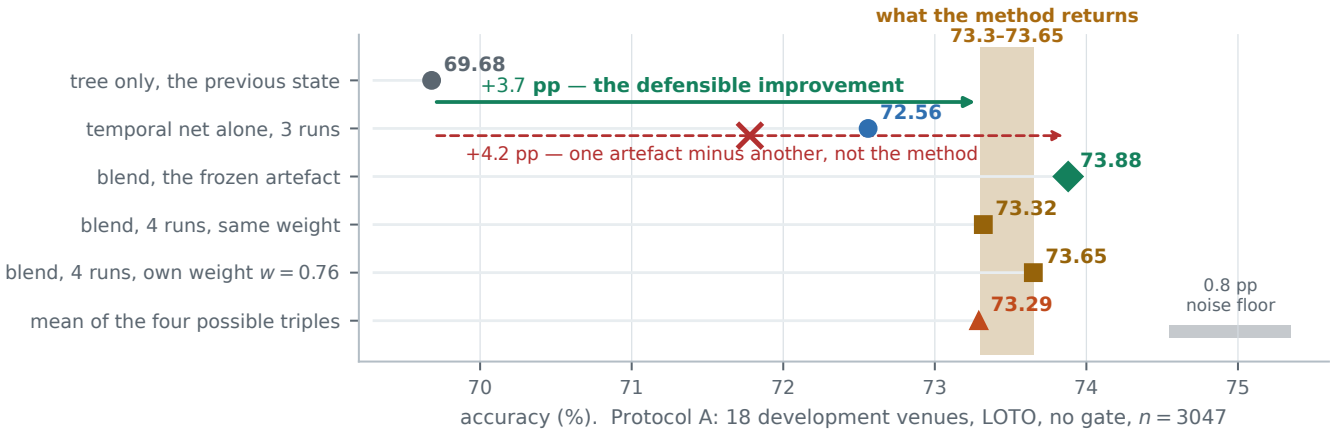


Figure 22. Protocol A in detail: the difference between the artefact and the method. The shaded band is what the method returns once a fourth run is measured. The solid arrow is the defensible improvement over the previous state; the dashed one is the number obtained by subtracting one artefact from another, which is not an improvement of the method.

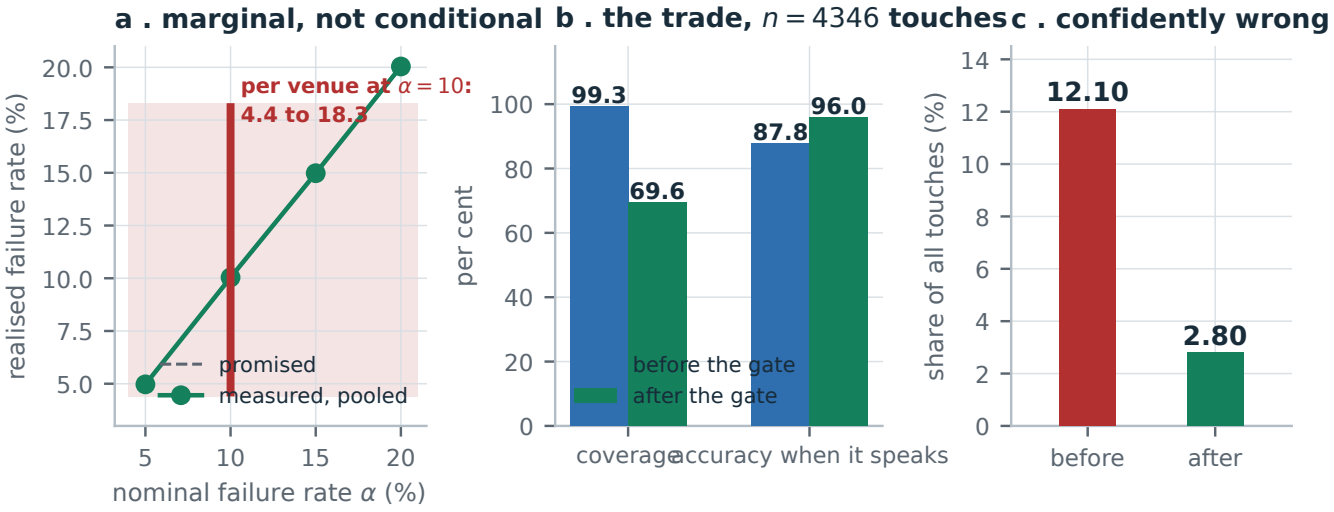


Figure 23. Protocol B. **(a)** The conformal guarantee is kept pooled and broken per venue: at $\alpha = 10\%$ the realised failure rate spans 4.4–18.3% across the 18 venues. **(b)** The trade on 4 346 held-out touches. **(c)** Confidently wrong verdicts as a share of all touches, before and after the gate.

5.3 Protocol C: what the acquisition failure costs

Protocol C asks a different question on a wider population: all 21 tournaments, 4 958 two-light pairs, 413 features, gradient-boosted trees [5, 8] with mirror augmentation and antisymmetric averaging, no gate, intervals by resampling whole tournaments with replacement over 4 000 repetitions.

Pooled accuracy is 68.43% [67.02, 69.81]. Split by whether the overlay reader works, it is 69.17% [67.76, 70.63] on the 16 readable venues ($n = 4217$) and 64.24% [62.81, 66.91] on the five blind ones ($n = 741$): a gap of 4.93 *pp* [2.14, 6.97] with $p(\leq 0) = 0.001$ (Figures 21(C) and 3(c)).

A second design confirms it. Holding out the actual blind pair gives +5.99 *pp* [+4.31, +9.88], $p = 0.000$. Holding out four pairs of *readable* venues instead gives +3.79, +3.21, −6.07 and −1.39 *pp*, mean −0.12 *pp*, and none of the four positively excludes zero. The gap is a property of the blind venues, not an artefact of holding two venues out.

How much of the gap is the anchor. About 1.4 of the 4.93 *pp* — somewhat more than a quarter — is attributable to the anchor. The supporting evidence is that the blind venues lose whole information channels rather than degrading smoothly: the clock-derived window source fires on 0 of 3 853 generic clips against 4 714 of 19 158 (24.6%) readable ones, and the white channel is read on 0.00% of generic clips against 8.96–9.36% of readable ones. The rest of the gap is not attributed, and per Section 2.3 it cannot be, because generalisation and acquisition are confounded in this split.

5.4 In what sense the number "would be more"

Three of these figures are higher than 73.88% or lower than it, and it is worth being explicit about which comparisons are legitimate.

There are exactly two senses in which the headline understates the system, and both come with a price.

1. **When the model speaks, it is right more of-**

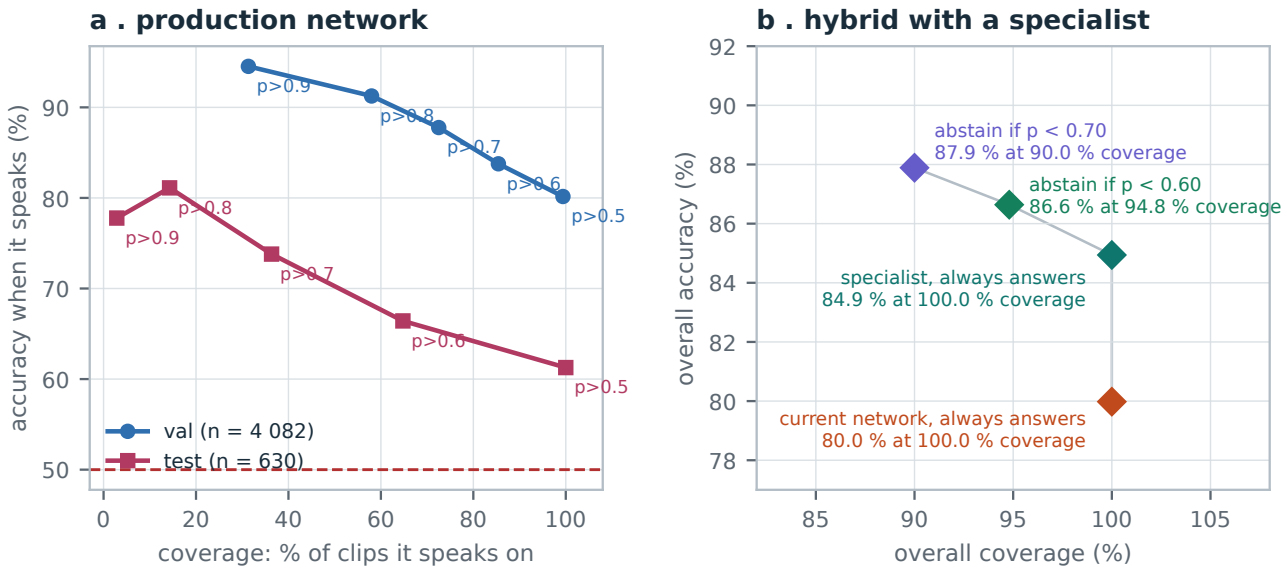


Figure 24. Risk against coverage for the production network and for a hybrid with a specialist head, at several abstention thresholds. Reused figure, re-rendered on white and in English. Every point on this curve is a different product, and choosing among them is a product decision rather than a modelling one.

ten. Under Protocol B the accuracy on answered touches is 80.5% on two-light touches, and the confidently wrong rate falls from 12.10% to 2.80%. The price is silence on roughly half the two-light cases, and the guarantee behind that number is marginal rather than per-venue.

2. **The pooled figure mixes venues where the data cannot be read.** Under Protocol C the five blind venues cost 4.93 *pp* of measured accuracy in those venues. That is a fault of data acquisition, it is diagnosed down to a normalised image band and four overlay geometries, and it is in principle fixable with five table entries — but until it is fixed it is part of the number.

What cannot be done is to take the highest of these, or a combination of them, and present it as "the real accuracy". Protocol B's 80.5% is conditional on answering and is not a floor for any venue. Protocol C's 69.17% is a different population and a different feature set from Protocol A's 73.88%, and the two must not be differenced. The system's honest description needs all three numbers and their protocols, and that is why they are reported three times rather than once.

6 Beyond priority

6.1 Action candidates

The same measurements support a coarse categorisation of what physically happened. Nine categories are assigned from pose and timing, and Figure 25(a) gives the distribution over 6 435 rows, with the scoring side agreeing with the manifest in 6435/6435. Two properties are deliberate: there is no catch-all, so "unknown" and "no action" both hold exactly zero rows, and locomotion is not a category — *advance* and *retreat* were folded back into the unclassified pool because nobody awards a touch "with an advance".

Coverage over clips that have pose archives rose from 12405/23018 (53.9%) to 18489/23018 (80.3%). The residual 19.7% is not a labelling backlog: it is clips whose role is genuinely indeterminate at the touch instant, and re-running the labeller on them returns indeterminate again. A full re-extraction at longer duration was costed at 15.5 GPU-hours and measured to yield zero new labels, so it was stopped.

κ measures reproducibility, not correctness. Figure 25(c) reports Cohen's κ [2] between two independent measurement paths — the production features and a second implementation using a 106-frame window and 29 columns — evaluated out-of-tournament with three seeds. The five current categories give $\kappa = 0.287$ ($n = 18549$) and a proposed three-box grouping gives $\kappa = 0.396$ ($n = 18838$). Every one of these has a within-tournament-shuffled twin at 0.000, so the agreement is not an artefact of the marginals.

It must be read for what it is. **There is no human-annotated reference set for this taxonomy anywhere in the project.** κ here measures whether two code paths that both read the same 2D poses agree with each other, which is a necessary condition for the categories to mean anything and not remotely a sufficient one. For calibration of expectations: a rule-based referee built from the rulebook identifies who scored 51.9% of the time. These are geometric candidates. They are not a referee's decision, and the categorisation is not validated as correct.

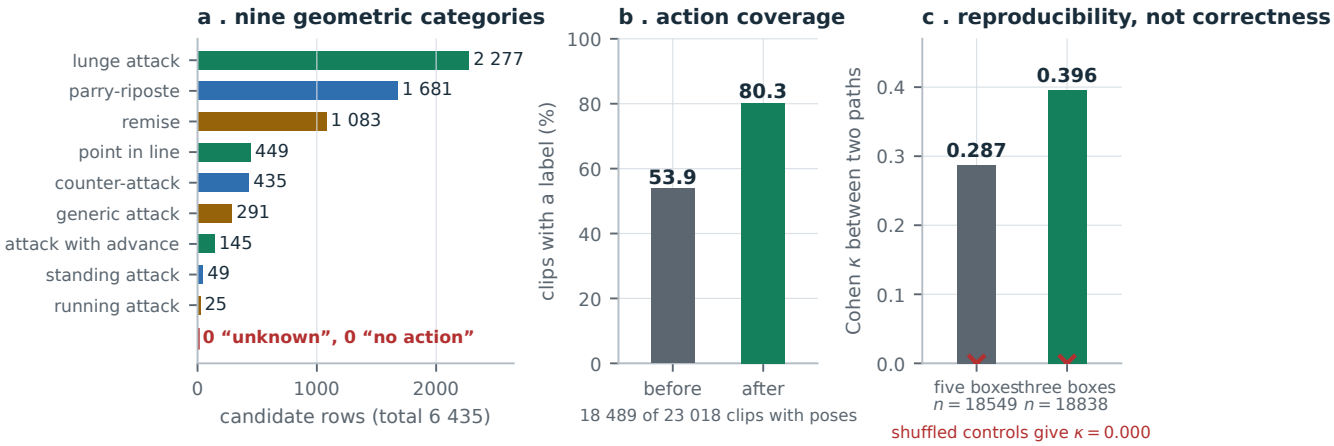


Figure 25. Action candidates. **(a)** The nine categories over 6 435 rows, with no catch-all: “unknown” and “no action” hold zero rows each. **(b)** Label coverage over clips with pose archives. **(c)** Agreement between two independent measurement paths. These are geometric candidates derived from pose and timing, never the decision of a human referee, and κ measures agreement between the two paths, not correctness: no human-annotated reference exists.

6.2 Per-fencer metrics, and which of them are reliable

A fencer profile aggregates the 68 measurements. Whether a given aggregate is a property of the fencer or of the clip was tested by split-half reliability over 142 fencers with at least six bouts and twelve clips per half, against a null in which the fencer label is permuted within venue. The result is a clean separation. Preparation steps ($r = 0.603$ against a null of 0.183), forward-step fraction (0.563 vs 0.187), cadence (0.546) and lunge p90 (0.310 vs 0.135) are fencer properties. Peak forward speed (0.408 against a null of 0.431), “starts before the opponent” (0.156 vs 0.169) and — most instructively — a fencer’s own touch rate (0.097 vs 0.096) are not: they do not repeat across halves any better than a shuffled label does.

There is a matching sample-size floor on head-to-head statistics. Split-half reliability by directed pair is 0.15 at three touches, 0.22 at five and 0.31 at eight, and a variance decomposition puts the minimum usable count between 19 and 46 touches depending on the category. The threshold was set at 20, and only 24 of 1 923 directed pairs reach it.

6.3 Stature

Body scale enters every normalised measurement, so stature matters even though it is not the product. It is estimated by a graph: fencers are nodes, clips are edges, each clip contributes the log ratio of the two fencers’ apparent statures, and 104 published heights anchor the system. The corpus graph has 401 nodes in a single connected component with a median degree of 58 clips per fencer, and the solution determines 297 statures that were not previously known.

The 104 anchors are heights published by third parties. They were not measured in this work, and nothing in this report should be read as an independent measurement of any individual’s height.

Validation is leave-one-anchor-out over the 104 anchors, refitting with the evaluated fencer excluded (Figure 26(a)): median absolute error 3.75 cm and p90 10.85 cm, against 6.75 cm and 14.00 cm for the sex-mean baseline it replaced. The paired difference is −2.01 cm with a bootstrap interval of $[-3.74, -0.83]$, and the graph wins on 68% of anchors. The regulariser is not doing the work: sweeping it from 5 to 1000 moves the error only from 3.62 to 4.29 cm.

The fit lies about its own uncertainty. This is the part that would be easiest to get wrong. The formal standard error the fit reports has a median of 1.40 cm, while the real median absolute error is 3.75 cm — understated by a factor of 2.7 — and its correlation with the real error is +0.11, so it does not even rank which estimates to distrust. The uncertainty that ships is therefore not the fit’s: it is a bar calibrated against measured error, with a floor of 3.81%, giving a median of 6.9 cm at one sigma. The unusable column is named `se_formal_cm_NO_USAR` in the output, which is the right way to retire a number.

How much sex-mean contamination the graph removes, and in what units. The graph removes a bias rather than merely improving an average, but the size of what it removes depends on how the question is asked, and three different measurements of it are easy to confuse. They are separated here because two of them are true at once and look contradictory.

The field that records what a clip *actually used* is the clip-level scale method. Before the graph it read a sex mean on 7 369 of the 23 011 clips (32.0%); with the graph it reads it on none. An independent audit of 1 200 artefacts drawn at random from the 23 011 materialised confirms the current state: the sex mean does not appear once, and the distribution is 69.1% from published records, 30.6% from the graph and 0.2% legacy, against the 67.6% and 32.0% that the corpus-wide table predicts for those two routes.

Counted *per side*, the exposure was much wider, and this is the number that matters for whether the scale itself was contaminated. The clip-level label reads sex

mean only when *both* fencers are on it, because the combined clip scale takes both heights and the label follows the side that has a published record. Before the graph, 7 304 clips (31.7%) had both fencers on the sex mean and a further 11 505 (50.0%) had exactly one of the two, so 18 809 clips (81.7%) carried a sex mean somewhere inside their combined scale while only about a third of them declared it. Both counts go to zero with the graph, because all 401 fencers resolve — 104 from published records and 297 from the graph — and no clip is left with a fencer without a stature.

The bias being removed is not symmetric. Where sex is not declared at all, which is the case in 10 256 of the 23 011 clips, the old fallback assigned 1.74 m to both fencers, inflating women's apparent scale and shrinking men's.

6.4 Perspective, scale and the piste

Apparent size is not size. Figure 27 makes the geometry explicit with a synthetic scene: two fencers of identical real stature, at 12 m and 26 m from a camera 6 m up and tilted 18°, differ in apparent height by a factor of 1.89 where the pinhole law predicts 2.00. Correcting for it is not optional: regressing apparent scale against ankle row over 2 796 clips gives a slope of -1.20 with $r = -0.851$, and after dividing by $(y - 0.14)$ the residual slope is -0.12 with $r = -0.162$ (Figure 28). The correction removes about nine tenths of the effect and does not remove all of it.

Even with the correction, per-clip scale is unstable in a way that bounds every metric distance in the system. Within a single clip the foot row moves enough for the inferred scale to change by 6.8% at the median and 21.4% at p90 between its own p10 and p90 (Figure 29).

The piste itself is the natural ruler, and it half works. The FIE piste is 14 m long with on-guard lines 2 m either side of the centre, so the regulation separation of those lines is exactly 4.00 m [3]. Measuring that separation through the system over 318 clips gives medians of 3.92 m and 3.90 m by two independent routes — a bias of -1.9% and -2.5%, and the two medians agree within 0.3σ of the standard error of the median (Figure 30). That is a genuine validation of the scale chain at the level of the corpus. It is not a validation per clip: the per-clip median absolute deviation is ± 1.16 m, and the production distance control on 305 clips reads 4.20 m, a $+5.1\%$ bias. Detecting the piste lines directly was tried and closed (Section 7.7).

7 Ablations and negative results

This is the section that decides whether the rest of the report is credible. Thirty-nine avenues were closed: thirty-eight built and measured against their own null control, and one, generalised cross-entropy, closed *by argument* and marked "not measured" in the ledger rather than quietly omitted, because a report of negative results should not hide the case it decided not to run. Table 5 is the ledger; Figure 31 plots twenty-seven of them against the 0.8 *pp* floor of Section 4.4.

The two counts in this report differ and each is verifiable by counting. The ledger has forty rows. One of them, stratified conformal prediction, is a partial rather than a closure: it fails at its stated purpose but buys a different guarantee the report keeps. That leaves **thirty-nine closed avenues**, and sub-families are counted as their own avenue because they carry their own reference, effect and null. The forest plot carries twenty-seven of them. The other thirteen are absent from the plot because their evidence is not a change in accuracy against the plot's common reference: a correlation, a detection rate, a single-venue figure, a label field that turns out to be empty, or an archive that contradicts itself on who scored.

Two conventions apply throughout. First, an avenue is closed when its measured effect does not clear the floor *or* does not beat its own null — both tests, not either. Second, several avenues produced a real effect that was closed anyway, for a stated reason; those are the interesting ones and they are marked.

7.1 Acquisition: audio, twice

Audio was attempted twice, on 100 two-light clips from the eleven development venues that have a sound track (Figure 32).

The first attempt tried to date the blade clash acoustically, to get a contact instant independent of the lamps. Timing resolution was never the obstacle: where a transient is visible at all, which is 91 of 100 clips, its full width at half maximum is 7.0 ms at the median and the instant can be fixed to about 2 ms. Identification was the obstacle. Using the buzzer as a known-time calibrator dates the touch to within ± 60 ms on 15% of clips against a chance rate of 12%. A hand-built metal-impact descriptor finds a candidate in the prior window on 59% of clips and in control windows on 41%, and beats the maximum of its own twelve controls on 16% against a chance rate of 8%. Simple spectral flux does separate the prior window from controls (17.2 against 13.8 dB, Wilcoxon $p = 0.0008$) but with an AUC of 0.625. The decisive test was a positive control: at a 10% false-alarm operating point, a guaranteed event is recovered on only 20–28% of clips. The pipeline works and the signal is not separable.

The second attempt tried the stereo image, on the hypothesis that panning encodes which fencer is where. Re-extracting with both channels retained, 77 of 100 clips are usable; two venues are effectively mono at -28.6 and -29.7 dB side-to-mid. The within-clip rank correlation between panorama and piste position is $+0.002$ ($p = 0.829$), the inter-channel delay is exactly zero in 84% of clips, and the four panorama descriptors give AUCs of 0.520, 0.481, 0.532 and 0.568 against a standard error of about 0.066. Again a positive control settles it: injecting a known positional amplitude difference, the pipeline recovers a correlation of 0.303 at ± 0.50 dB and 0.171 at ± 0.25 dB. Measuring 0.002 therefore bounds any real positional component below ± 0.2 dB.

Both attempts also face a structural blocker: 1 316

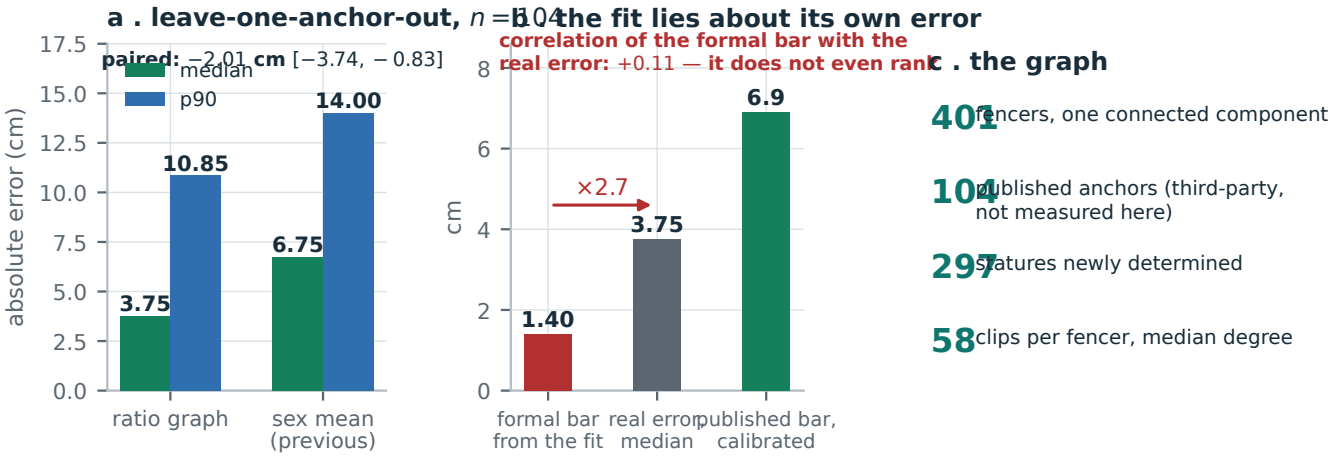


Figure 26. Stature by ratio graph. **(a)** Leave-one-anchor-out over the 104 published anchors, against the sex-mean baseline. **(b)** The fit's own standard error understates the real error by 2.7× and correlates with it at only +0.11; the published bar is calibrated against measured error instead. **(c)** Graph structure. Stature here is an estimate with an uncertainty, not a measurement, and the anchors are third-party published values.

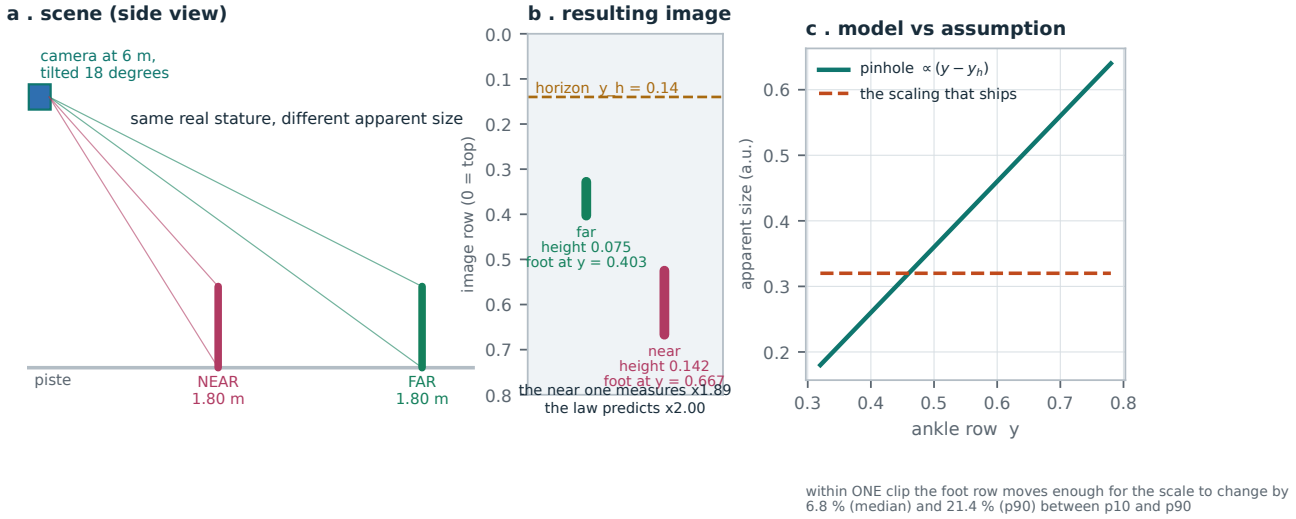


Figure 27. Why apparent size cannot be used as size, on a synthetic scene with the camera geometry stated. Reused figure, re-rendered on white and in English.

of the 3 130 development two-light clips have no audio stream at source, 42%, clustered in eight whole venues. Even a working audio feature would have been unavailable exactly where venue robustness is tested.

7.2 Temporal: window length, adaptive windows, long context

Window length. Micro-scale features over a 1.2 s window score 65.97% as a block on their own and add +1.06 *pp* [+0.24, +1.74] on top of the 413-feature reference, which clears the floor and was kept. Macro-scale summaries over 4.0 s add +0.22 *pp* [−0.25, +0.64] on top of that and were not. A flat 4.0 s window at the same resolution is the only significantly *negative* contrast in the sweep, −1.61 *pp* [−3.13, −0.23].

Adaptive window. The motivation was solid and was verified: the commitment point is detectable in 3 033 of 3 058 clips, the phrase has a median length of 1.30 s, and a fixed 1.2 s window therefore measures dead time in 42.6% of clips and truncates in 57.2%. Adapting the window to the commitment point gives +1.63 *pp* on the 413-feature block — and **a duration drawn at random from the same empirical distribution gives** +1.60 *pp*. The whole effect is the presence of a per-clip duration column, not the correctness of that duration. Over a reference that already contains micro features, the real version gives +0.10 *pp* and the fake gives +0.00. The best configuration in the whole directory is +0.19 *pp* [−0.62, +1.14]. Closed.

Trimming the window. Trimming to 2.3 s to buy throughput was measured on 500 re-extracted clips and closed for analysis, with a warning worth generalising (Figure 33). 402 of the 413 features change value: 236 break ($r < 0.90$), 16 become non-computable, and all twelve on-guard features break with a median r of 0.434. Worse, of the 39 features with a nominal 4.0 s window, the 4.0 s value *equals* the 2.0 s value in 91.9% of compar-

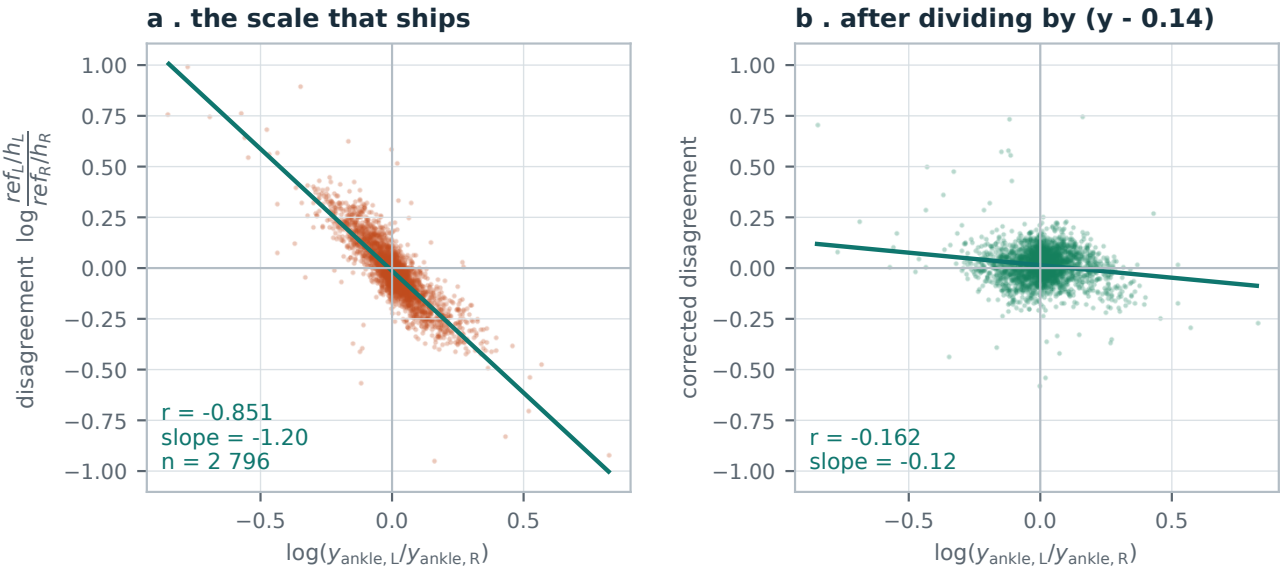


Figure 28. The perspective correction, measured on 2 796 real clips: the raw dependence of apparent scale on image row, and what is left after dividing by $(y - 0.14)$. Reused figure, re-rendered on white and in English.

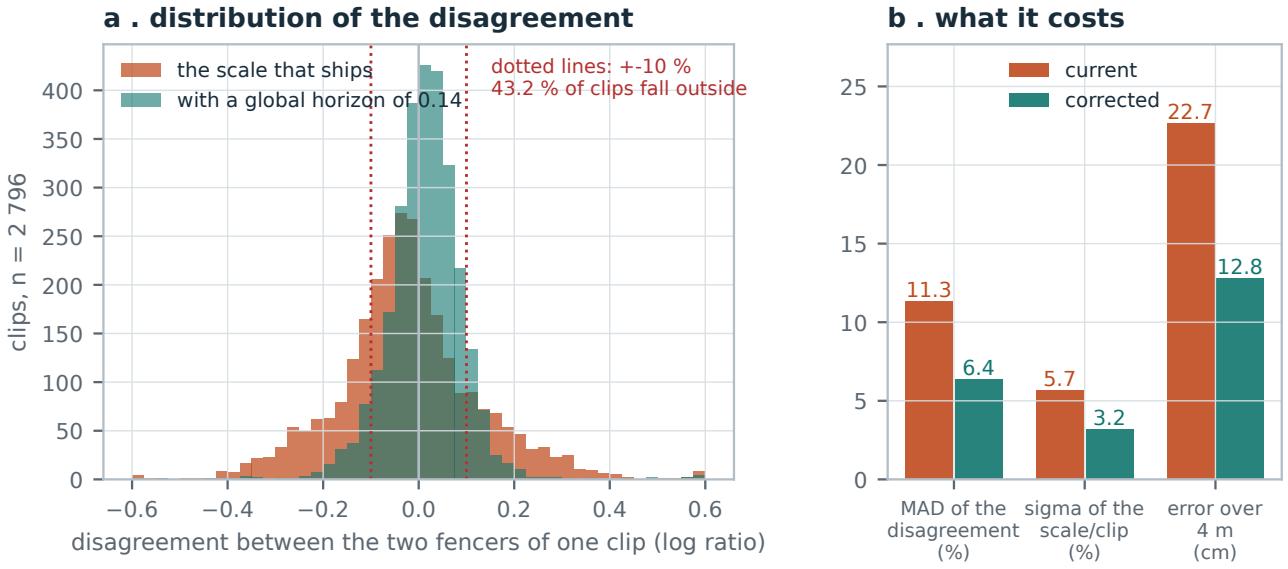


Figure 29. Disagreement between the two fencers of the same clip, before and after the correction, and the residual instability of scale within a single clip. Reused figure, re-rendered on white and in English.

isons — a column can keep a perfectly plausible value and quietly stop meaning what its name says. The control that ties the change to the window and nothing else is clean: on the 56 clips that lost zero frames, the change in predicted probability is exactly 0.0000 and no decision moves.

Long context. Going the other way, re-extracting 1 104 clips at full duration and using all 413 features over the whole clip gives $-2.45\,pp$ $[-5.17, +0.61]$, and the two null-by-construction blocks give $+0.39$ and $+1.08\,pp$ — adding information-free columns beats adding the real ones. The context sweep is flat from 1.5 to 14.1 s. The reason is measurable: the decisive episode starts at a median of -0.60 s and falls outside the 4.3 s window in only 1.4% of clips. There is very little to gain because there is very little out there.

7.3 The apparatus as a predictor

Inter-lamp delay as a model input. Adding the inter-lamp delay as a feature costs $-0.68\,pp$ $[-1.67, +0.50]$ on the three-seed ensemble — and **a shuffled version of the same column costs** $-0.65\,pp$, which is the same number. The column carries nothing the model does not already have: the delay is $R^2 = 0.53$ predictable from the eight lamp channels alone, with the sign right 89% of the time. Section 1 explains why it could not have helped much even if it were new infor-

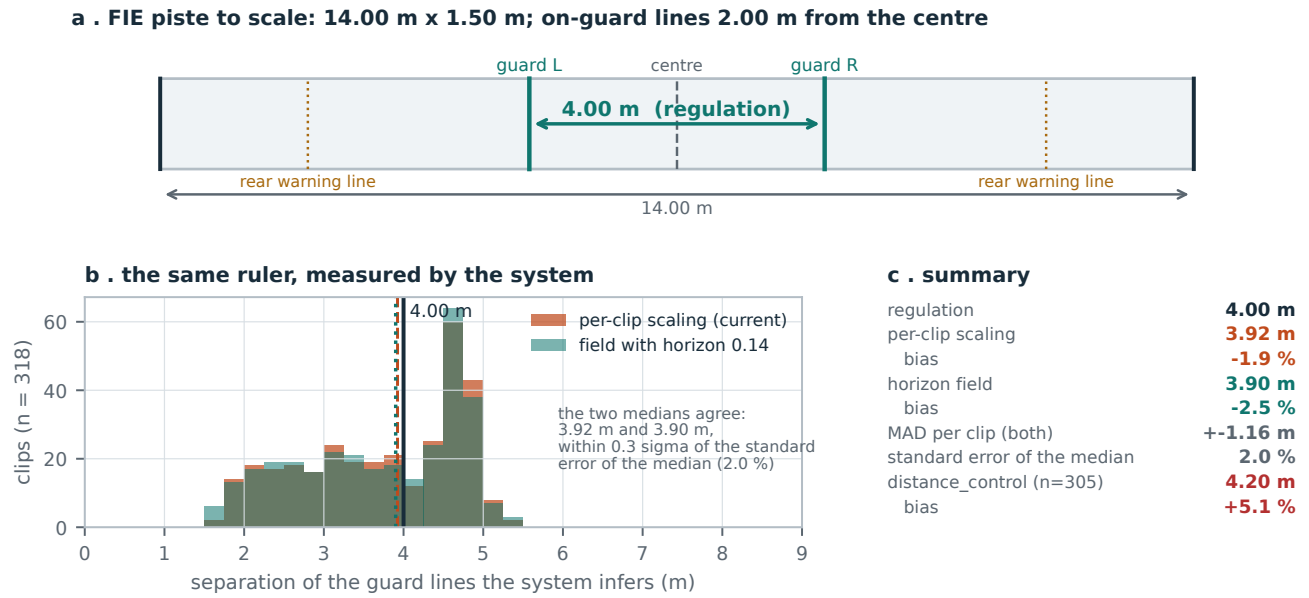


Figure 30. The piste as a ruler. The regulation separation of the on-guard lines is 4.00 m; the system infers medians of 3.92 and 3.90 m over 318 clips, with a per-clip median absolute deviation of ±1.16 m. The chain is right in aggregate and loose per clip. Reused figure, re-rendered on white and in English.

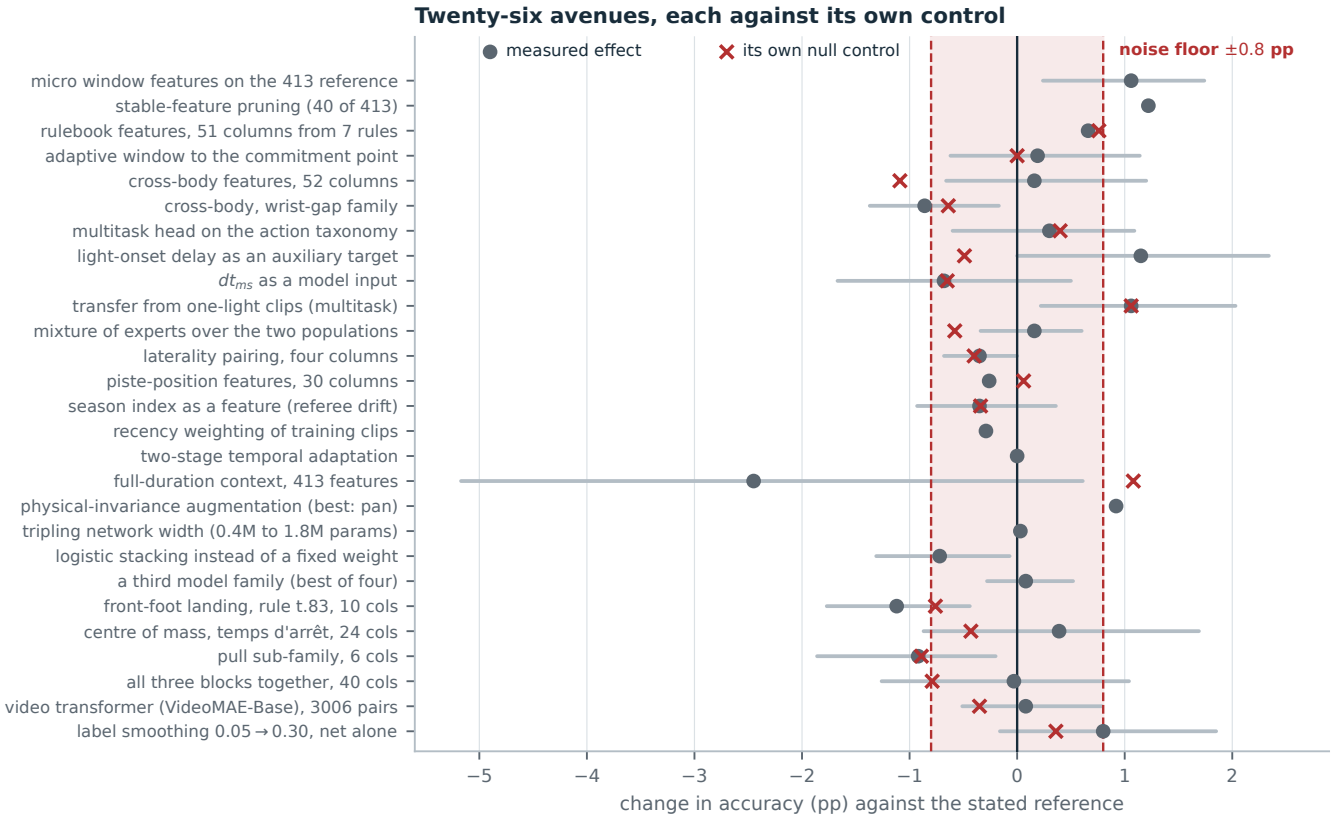


Figure 31. Twenty-seven avenues, each against its own null control. Circles are measured effects with tournament-bootstrap intervals where available; crosses are the same construction run on a control that cannot carry information. The shaded band is the 0.8 *pp* noise floor. An avenue whose cross sits at or beyond its circle has not been shown to work, whatever its point estimate. Effects are against the reference stated in Table 5; because the references differ, the rows are not comparable with one another as accuracies.

mation.

Inter-lamp delay as an auxiliary target. The same quantity used as an auxiliary regression target does work: $+1.15\,pp$ $[+0.00, +2.34]$ on the ensemble against $-0.49\,pp$ for its shuffled control, and the blend rises from 73.55 to 74.37%. It was closed anyway, for a reason that is a good example of why venue robustness

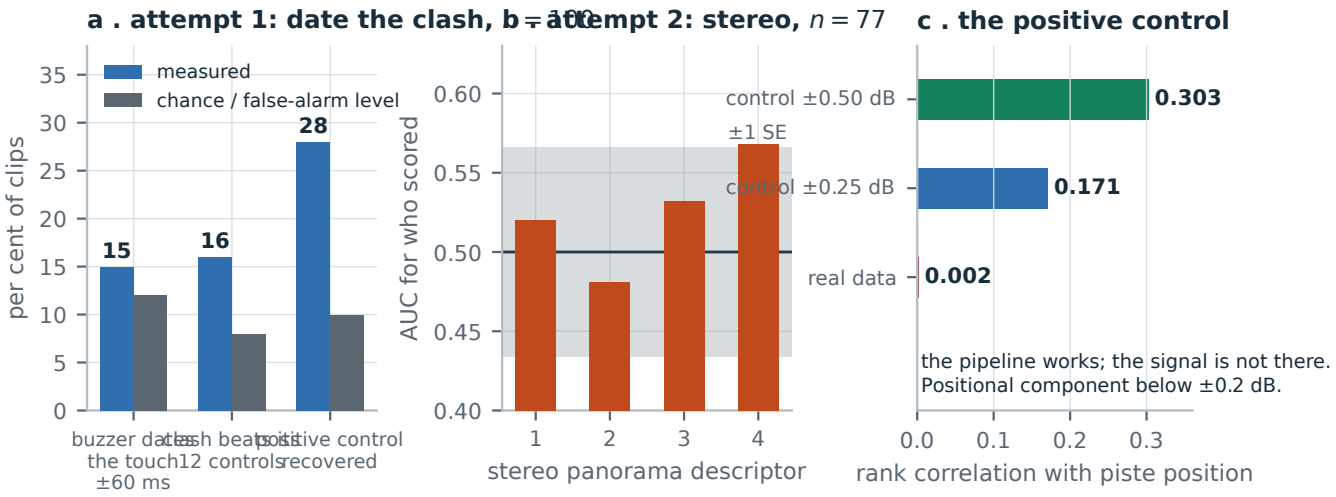


Figure 32. Audio, two attempts, both closed. **(a)** Dating the blade clash: every measured rate sits within a few points of its own chance rate, and a guaranteed event is recovered on 20–28% of clips at a 10% false-alarm operating point. **(b)** Stereo panorama descriptors against who scored; the band is ±1 standard error around chance. **(c)** The positive control that closes it: the pipeline recovers an injected effect, so the absence of a measured one bounds the real effect rather than the method.

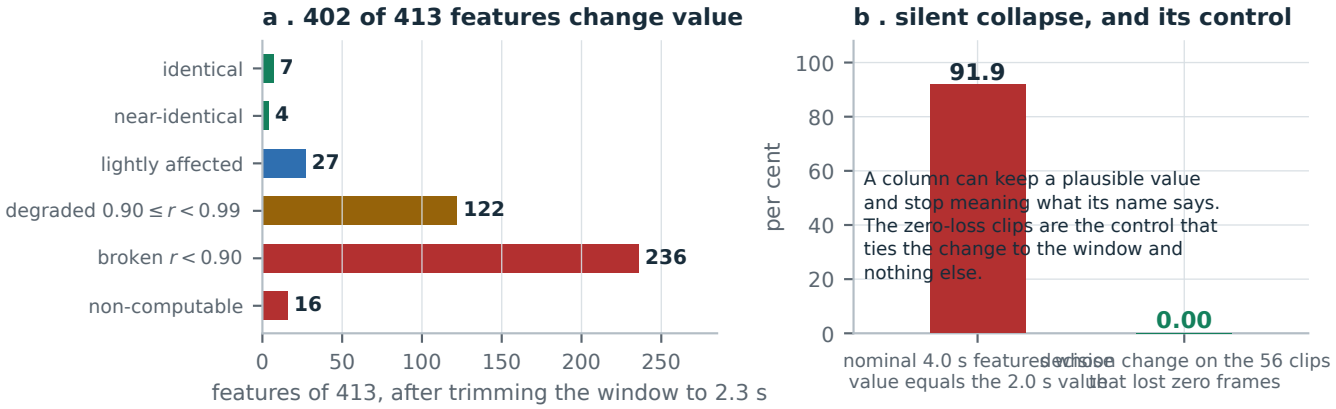


Figure 33. Trimming the window, and why a feature can lie quietly. **(a)** 402 of 413 features change value when the window is trimmed to 2.3 s; 236 break outright. **(b)** The silent-collapse rate, and the zero-frame-loss control that attributes the change to the window and nothing else.

has to be checked separately from accuracy: the gain is +1.32 *pp* on the 2 661 clips from readable venues and +0.00 *pp* on the 386 from generic ones, and *both held-out tournaments are* 100% *generic*. The feature improves the model exactly where the model is already fine, and does nothing where it is tested.

Simultaneity as an abstention signal. Two lamps in the same frame account for 15.6% of touches, and accuracy on them is 72.07% — above average, not below. Model confidence is a better abstention signal than any simultaneity taxonomy by about 14 points, so the simultaneity feature ships disabled.

7.4 Feature families

Rulebook features. Fifty-one features were derived from seven articles of the rulebook. As a block they give +0.66 *pp* — and their own time-shifted null gives +0.76 *pp*. The block does not beat its own control, and only 5 of the 51 pass the stability criterion. Two occurrence measurements explain why the rulebook does not translate directly: a point in line is established in 0.60% of fencers (14 of 2 315), and loosening the angle threshold from 155° to 170° moves that only between 2.81% and 0.17%. Meanwhile a foot-to-lamp delay sweep across 80–320 ms gives AUCs of 0.487–0.498 over 2 861 pairs: flat, and on the wrong side of chance.

Cross-body features. Fifty-two features describing the geometry between the two fencers give +0.16 *pp* [−0.66, +1.20] as a block. The wrist-gap family within it is worse than its own nulls at −0.86 *pp* [−1.37, −0.17]. The cause is redundancy, not weakness: the best new feature correlates at 0.979 with an existing one, and the best single univariate discriminator among them reaches 0.5699 against 0.6750 for a hip-initiative feature already in the reference. One clean positive survives — composing four raw conditions into six state features is worth +0.67 *pp* over the raw conditions at equal column count — which is a statement about representation, not about new information.

Piste-position features. Thirty features describing where on the piste the action happens give $-0.26\,pp$, below both their shifted null (+0.06) and their phase-randomised null (0.00), and below both noised versions of themselves (+0.57 and +0.54 at two noise levels). When perturbing features with Gaussian noise improves them by half a point, the whole range is fitting noise. Univariate AUCs are respectable at 0.607–0.658, so again the failure is redundancy: correlations up to 0.969 with features already present. Coverage compounds it — absolute piste position exists for only 10.5% of development clips.

Laterality pairing. Handedness is the cleanest variable in the project: read in 3 049 of 3 130 clips, agreeing with the federation record on 96.9% of 295 fencers and 100.0% of the 84 declared left-handers, and stable under keypoint noise in 99.84% of fencers. It still fails. Four laterality columns give $-0.35\,pp$ $[-0.68, +0.00]$; explicit interactions give $-0.73\,pp$; separate models per pairing lose $1.98\,pp$, all of which is sample size. And there is a structural reason it cannot work as a marginal feature: the mirror of Equation 1 maps right-right bouts exactly onto left-left bouts, so a symmetric laterality variable has no marginal signal by construction.

What is real is an interaction that cannot be spent. Armed-hand height discriminates at AUC 0.524 in homogeneous bouts and 0.615 in mixed ones, a difference of $+0.091$ $[+0.043, +0.139]$ with permutation $p < 0.001$ across 10 of 13 venues; and mixed bouts are simply easier, 73.23% against 70.26% and 69.96%. The finding is solid and worth less than $0.8\,pp$.

7.5 Populations

Transfer from one-light clips. There are 11 460 one-light development clips against 3 130 two-light ones, so transfer is the obvious lever. Every route was measured with a shuffled-label twin, and the twins are devastating: multitask transfer gives $+1.06\,pp$ $[+0.22, +2.03]$ and multitask transfer *with labels shuffled within tournament* gives $+1.06\,pp$; pre-train-and-fine-tune gives $-0.26\,pp$ while pre-training on deliberately false labels gives $+0.51\,pp$. Lying is better than telling the truth, which means the effect is regularisation and not knowledge.

It is also not the same task. A model trained only on one-light clips and evaluated on two-light clips scores 49.32% — chance. The rank correlation of univariate feature strength between the two populations is −0.115, and only 44.3% of features share the sign of their effect. And one-light is the harder problem despite being the larger: 55.24% with 11 414 clips against 63.26% with 3 120, still 56.24% when subsampled to equal size.

No-point and correction clips. Neither can be learned here, for a reason of construction rather than difficulty: the label field is left or right in all 16 933 index records, so no "no point" class exists to train or validate on. Clips flagged as corrections are excluded from both the strict and the lax criteria because the archive contradicts itself about who scored them; in one worked matchup, 3 of 21 clips are corrections and remain unusable. The gap between what can be asserted and what can merely be shown is large: the strict criterion admits 11 797 touches of 34 374 and the lax one 29 867.

Competition level. Major championships look harder — 69.76% against 71.73% under the frozen block — but the effect depends entirely on where the class boundary is drawn. Taking three championships gives $-1.3\,pp$ with $t = -0.37$; taking only the world and European championships gives $-4.6\,pp$ with $t = -2.81$. An effect that changes by a factor of three with the definition of its own predictor is a confounder to control for, not a feature to add.

Referee-criterion drift. Ten variants of the hypothesis that refereeing has drifted across seasons were measured, and it survived none. The premise is false in both halves of the data: 58.0% of development pairs are from 2022 or later, while both held-out tournaments are older. Accuracy against season gives Spearman −0.030 and a slope of $-0.09\,pp$ per season with a standard error of 0.74. Exponential recency weighting is at best $0.29\,pp$ *below* doing nothing, and against a shuffled-season control the real weighting sits at the 62nd percentile of the null. Season as a feature has infinite venue footprint — zero within-tournament variance — and a placebo in which season is permuted across venues performs identically to the real thing (-0.34 against $-0.35\,pp$). Two-stage adaptation gives $+0.00\,pp$. A direct data diagnostic that does not pass through the classifier agrees: the bent-arm-during-advance disadvantage is −11 to $-16\,pp$ in all six seasons, with a slope of $+0.124\,pp$ per season against a standard error of 1.477, and it is negative at 16 of 18 venues.

One circular route was found and closed explicitly: dropping old bent-arm wins would delete 19.7% of pairs and drive the old-era win rate to exactly 0.0%, manufacturing a +60 to $+70\,pp$ "drift" out of nothing.

7.6 Models and combination

A third model family. Four were built and all four were rejected by the same arithmetic. A graph network over the two skeletons (34 nodes, 98 cross edges, 468 990 parameters) scores 68.69%, correlates with the temporal net at 0.653, and receives weight exactly 0.00 — it would need 71.7% to earn any. A regularised linear model scores 68.26% and needs 73.5%. Neighbour-based classification scores 64.59%, earns weight 0.20 and contributes $+0.08\,pp$ $[-0.28, +0.52]$. A rule engine built from the rulebook scores 56.12%, answering on 85.8% of two-light touches and being right on 56.6% of those.

How to combine. The fixed weight is not beaten by anything more sophisticated: logistic stacking $-0.72\,pp$ $[-1.31, -0.07]$, gradient-forest stacking $-0.83\,pp$, rank averaging 73.29%, confidence weighting $-0.75\,pp$, local-competence weighting between -0.30 and $-0.65\,pp$, and

a free non-parametric fit on the two-probability plane between 73.35 and 73.38%. All below 73.88%.

The blend weight, and why the gate has to be in the loop. Raising the weight from 0.65 to 0.85 looked right: ungated, it buys +3.76 *pp* at the worst venue. **Measured with the gate recalibrated for each weight, the conclusion inverts** (Figure 34): 0.65 gives 73.68% at the worst venue and 0.85 gives 69.44%, with three points less coverage. The mechanism is legible — raising the weight drops the Platt slope from 1.030 to 0.845, so the gate must harden to deliver the same guarantee, and it ends up covering less and being right less where it hurts. The location of the optimum is itself noise: the argmax jumps between 0.55 and 0.95 across guarantee levels, and the plateau within the 0.8 *pp* floor spans 0.40–0.93. What is not noise is that high weights are worse in every set tested. Any future proposal to raise it has to be measured *with* the gate.

Extra runs, and capacity. Adding a fourth run leaves the network a tenth of a point better and the blend average worse. Marginal gains per run are +0.56 then +0.38 *pp*, so going from three to six runs is worth +0.4 to +0.5 *pp* — below the floor — at +33% inference cost each. More runs narrow the estimate; they do not raise it. Capacity is the same story: widening the network across 96, 136, 176 and 208 units, which takes it from 0.4 M to 1.8 M parameters, moves accuracy by +0.03 *pp*.

Physical-invariance augmentation. Four transforms across three feature blocks and 18 venues. The best is a pan augmentation at +0.92 *pp*, with a paired t of 1.74 and wins at 9 of 18 venues. Nothing reaches two standard errors. Closed.

Multitask on the action taxonomy. Predicting the action category as an auxiliary task gives +0.30 *pp* [−0.60, +1.09] against +0.40 *pp* for labels shuffled within tournament. The head is not broken — it reaches 63.1% against a 60.8% majority baseline with real labels and 59.5% with shuffled ones — but the label quality bounds it: a second independent measurement path agrees at $\kappa = 0.205$ over 2 491 pairs, and the two largest categories are indistinguishable from each other at AUC 0.586 over 1 438 clips. The diversity argument also fails: correlation with the tree *rises* from 0.651 to 0.660.

7.7 Geometry: monocular depth and piste lines

Monocular depth. Four monocular depth models were run over 500 on-guard frames and 100 clips at 30 frames each (Figure 35). Depth *ordering* works: which fencer is nearer is recovered on 95.2% of frames once the two feet differ by 0.02 in image row, and on 100% at 0.05. Metric *scale* does not. No head emits intrinsics, the two variants of the same model disagree by a factor of 2.5

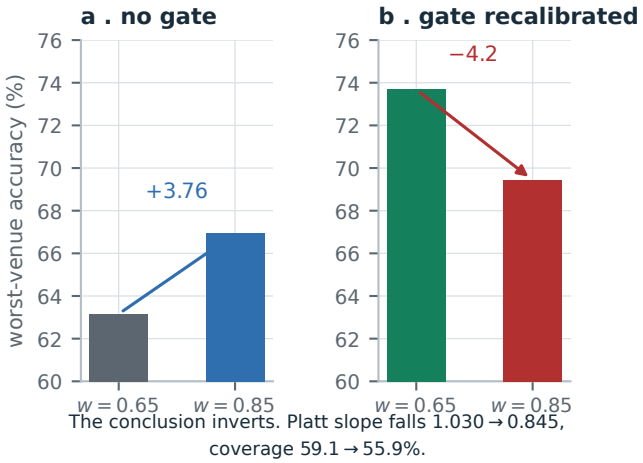


Figure 34. The blend weight, measured without and with the abstention gate in the loop. Both panels report accuracy at the worst venue, which is the quantity a single-competition product is exposed to. Measuring without the gate reverses the decision.

on the same image, and measuring the known 4.00 m on-guard separation gives a per-frame median absolute deviation of 46–51% under any focal-length convention; a focal length forced to produce 4.00 m implies a 64–113° horizontal field of view. Stature through depth lands at 9.0% median error against 2.1% for the ratio graph of Section 6.3. Reconstructed hand depth is worse still: the wrist-to-elbow depth difference is 33–126% of the projected forearm length and correlates with foreshortening at +0.010 to +0.063 over 1 672 forearms.

The one apparent gain — lower between-venue dispersion, 6.8% against 13.2% for the guard-line ruler — does not survive its own control. The per-venue biases of the two depth models *anticorrelate* at $r = -0.122$ across 16 venues, so the bias is a property of the model and not of the venue, and a leave-one-venue-out correction makes the better model worse (12.3 → 14.1%). Cost for the record: 1.3 GPU-hours for one frame per clip, 38.4 hours for thirty, 288 hours with the most accurate head.

Piste line detection. Detecting the piste markings directly was measured on 500 held-out on-guard frames across 19 venues and returns a calibration on 19 of them — 3.8% [2.4, 5.9] — abstaining on the other 481 (Figure 36). The decisive control is the human ceiling on the same rectified strips: 1 of 23, 4.3% [0.8, 21.0]. The detector is already at the ceiling of the evidence, so this is a data problem and not an algorithm problem.

The cause is geometry. A television frame spans a median of 8.3 m of piste (p05 5.8, p95 11.2, $n = 348$) and *none* of the 348 frames spans the full 14 m, while the piste carries only three usable marks. Format is not the bottleneck either: native video, rescaled video, JPEG and a temporal median plate all land between 1.2% and 8.5%. And the failure mode is the one that matters most — the band latches onto advertising hoardings in 6 of 23 strips, 26%. Against the deployed body-scale ruler on 12 clips it is biased +3.8% with a median absolute deviation of 13.1%.

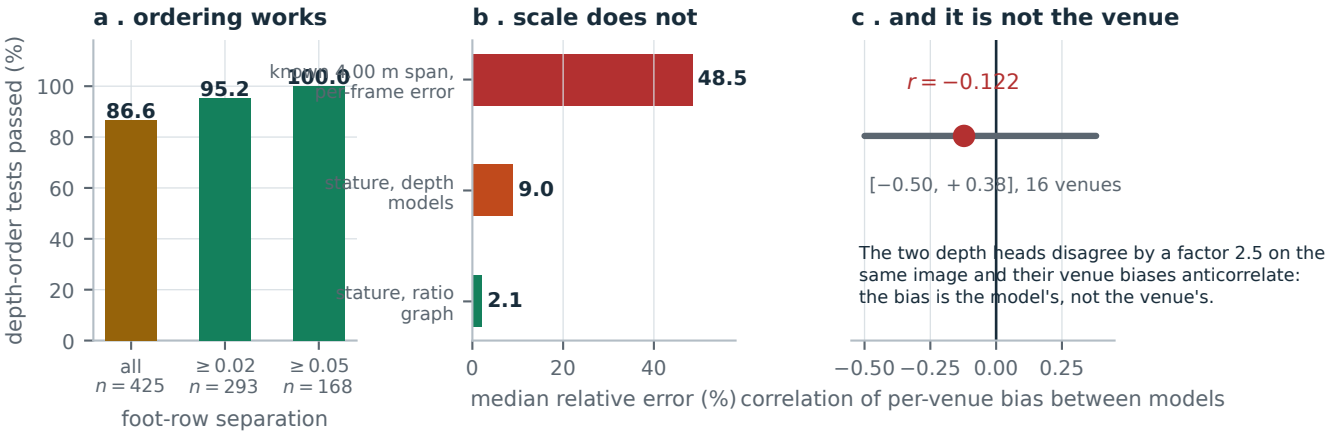


Figure 35. Monocular depth: ordering yes, scale no. **(a)** Depth-order tests pass once the two feet are separated in image row. **(b)** Metric error on a known 4.00 m span, on stature through depth, and on stature through the ratio graph. **(c)** The per-venue biases of the two depth models anticorrelate, so the bias belongs to the model rather than the venue.

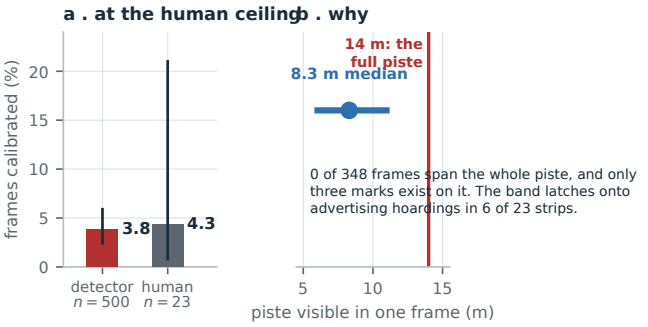


Figure 36. Piste line detection. **(a)** The detector calibrates 3.8% of frames and an assisted human reviewer calibrates 4.3% of the same rectified strips: the detector is at the ceiling of the evidence. **(b)** Why — the frame never contains the whole piste, and the marks that would disambiguate it are outside it.

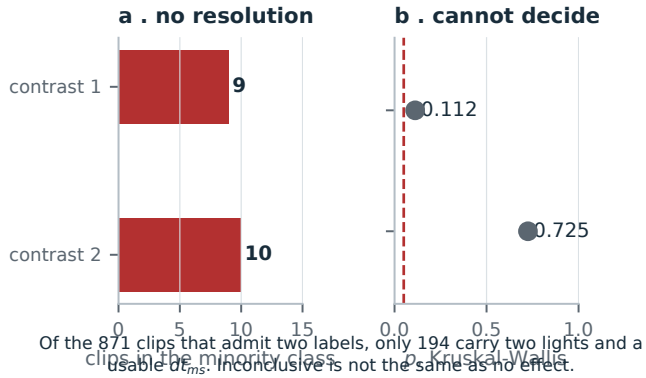


Figure 37. The 871 clips that admit two labels. **(a)** With 9 and 10 clips in the minority class the contrast has no resolution. **(b)** Both p-values. The avenue is recorded as inconclusive, which is not the same as recording no effect.

7.8 Labels

The 871 clips that admit two labels. 871 clips carry both a body-descriptor action type and an arbitral-role box, and the two disagree. An attempt was made to decide between them by contrasting each against the apparatus's temporal advantage on doubles, and **the test cannot decide** (Figure 37): only 194 of the 871 have two lamps and a usable inter-lamp delay, the two Kruskal–Wallis contrasts give $p = 0.112$ and $p = 0.725$ with 9 and 10 clips respectively in the minority class, and the AUC difference of 0.053 is within noise. The 871 stay as they are, labelled as measuring two different things. This is an inconclusive result, not evidence of no difference, and the distinction is the point.

How much label noise there is. Flipping the 19 bouts (0.61%) whose label the detector disputes gives $+0.28\,pp$; flipping the 75 bouts (2.40%) disputed by a stricter evaluation gives $-0.84\,pp$. Injecting noise deliberately calibrates the scale: 5% noise costs $1.9\,pp$ and 16% costs $8.4\,pp$. Against that scale, the measured label disagreement rate — lamp against annotation over 11 460 one-light clips — is 4.28% [3.92, 4.66], which puts a label-noise ceiling at about 95.7%. The same reports estimate an information ceiling of 65–67% on the features available, which is a useful thing to keep next to the accuracies in Section 5.

7.9 Data augmentation

Listed for completeness and marked as unquantified. Twenty transformations per profile were applied to two real training clips; the conservative profile retained 97.5% of usable frames and an aggressive, diagnostic-only profile retained 35.8%, with five of its twenty variants returning no usable frames at all. Those percentages describe how many pose observations survive a transform. They are not accuracy, $n = 2$ clips, and **no accuracy consequence was ever measured**, so this avenue is reported as attempted and not quantified.

7.10 An effect that clears every bar and still does not count

This subsection is the most instructive closure in the report, because the avenue passed the noise floor, passed a Bonferroni-corrected significance threshold, passed three seeds and a degeneracy check, and **still should**

not change the product. It is kept at length because the three reasons it fails are reasons that would not have been caught by any of the tests the rest of this report relies on.

The commission was thirteen arms of robust losses — generalised cross-entropy and an antisymmetric latent — against the premise that referee label noise was costing accuracy. That premise had already been refuted elsewhere in this report. Before spending the thirteen arms, the cheaper neighbour was swept first: the label smoothing that the network had been carrying, frozen at 0.05, never varied.

It sits at the minimum of its own curve. Paired clip-by-clip against the deployed value, with the deterministic backend of Section 4.4 and tournament resampling:

smoothing	accuracy	$\Delta\,pp$	p	ECE
0.00	70.40%	+0.62	0.074	—
0.05 *(deployed)*	69.77%	—	—	0.1137
0.20	70.53%	+0.75	0.067	0.0446
0.30	71.15%	**+1.38**	0.015	**0.0215**
0.40	71.28%	+1.51	0.014	0.0398
0.50	71.41%	+1.64	0.002	0.0747
kernel null	70.13%	+0.36	0.235	—

At three seeds, 0.30 returns **+1.03 pp** [+0.29, +1.81], $p = 0.003$, with 571 clips decided differently. It clears all three bars: the 0.8 *pp* floor; Bonferroni over the five arms first planned, 0.010; and Bonferroni over the seven actually run once the grid was extended, 0.0071. The effect shrank from +1.38 at one seed to +1.03 at three, exactly as the single-seed caveat predicted, and it is **not homogeneous**: +1.38, +1.21 and +0.49 by seed.

Why 0.30 **and not the** 0.50 **that scores higher.** Accuracy is weakly sensitive to logit scale and keeps rising to the edge of the grid, which is the signature of an unbounded optimum and a poor basis for choosing. Calibration is directly sensitive and **has an interior minimum at** 0.30: expected calibration error falls from 0.1137 at the deployed value to 0.0215, a factor of five, then rises again at 0.40 and 0.50. Choosing on the sensitive metric with an interior optimum is the defensible move; choosing on the insensitive one at a grid boundary is not. A degeneracy check rules out the obvious alternative explanation: mean p stays at 0.508 and median $|p - 0.5|$ at 0.206, so the gain is not a collapse toward the threshold.

The mechanism is consistent with both facts at once. Accuracy gains come from borderline clips crossing to the correct side once excess confidence is removed, and that is exactly what a better-calibrated head does without needing the decision boundary to move much.

Reason one: the estimand was the wrong one, and it cost the interval. Two conventions exist for pooling three seeds and they are not the same quantity. The ablation harness averages the *per-clip accuracy* of three models and then subtracts, which is what gave +1.03 *pp* [+0.29, +1.81]. The deployed artefact averages the three *probabilities* and thresholds once, which is an ensemble rather than a mean of models. Measured under the deployed convention the same comparison returns **+0.80 pp** [−0.16, +1.85] — **the interval no longer excludes zero**. "Mean accuracy of three models" and "accuracy of the ensemble of their averaged probabilities" are different estimands, and only the second is the product. As a pipeline check, the deterministic baseline scores 72.07% as an ensemble against 72.56% archived, consistent with the per-seed kernel drop partially averaging out.

Reason two: the blend damps it to a third. What matters is the artefact, not the network:

comparison (0.30 minus deployed)	$\Delta\,pp$	95 % interval
network alone	+0.80	[−0.16, +1.85]
network alone, with Platt	+0.80	[−0.16, +1.85]
blend at 0.65, uncalibrated	+0.24	[−0.53, +1.02]
blend at 0.65, with Platt	+0.38	[−0.24, +1.06]

At weight 0.65 against a tree that is unaffected, and with an inter-head correlation of 0.651, +0.80 on one component becomes +0.24 to +0.38 on the artefact: below the floor, interval crossing zero.

Reason three: the calibration gain was already being collected elsewhere. The fivefold ECE improvement was real but measured in the wrong regime — one seed, uncalibrated, which is the worst configuration rather than the deployed one. Under the deployed pipeline, averaging three seeds *already* removes most of the overconfidence (0.1137 → 0.0603) and the Platt stage that already exists removes the rest (0.0603 → 0.0258). The remaining advantage of 0.30 is 0.0258 against 0.0225: a factor of **1.15**, not five.

Worse for the hypothesis, the fitted Platt slope is 0.612 for the deployed value, meaning logits must be *shrunk* because confidence is excessive, and **1.209** for 0.30, meaning they must be *expanded* because confidence is now insufficient. **Smoothing at** 0.30 **overshoots**: it converts overconfidence into underconfidence. Whatever the calibration optimum is under the real pipeline, it lies between 0.05 and 0.30 and not at 0.30, and the "interior optimum" identified earlier belonged to the single-seed uncalibrated regime as well. The abstention gate confirms the same: coverage rises slightly at both thresholds with accuracy marginally worse at 0.60 and marginally better at 0.70, mixed and small.

Verdict, and what survives. The change is not adopted. The avenue is closed for a reason none of this report's other closures would have found: the effect exists in the network and does not survive the weight of the blend, which is where it has to survive to be worth anything. What remains true and is new: a hyperparameter carried frozen at 0.05, never once varied, sits at the minimum of its own accuracy curve, and the network is measurably better at 0.30 under the ablation

convention. What does not follow: any change to the deployed artefact. The population throughout is 3 059 pairs over 19 venues, $n = 3\,047$, which is what the network can measure and not the 4 958 pairs of the tabular protocol.

The transferable lesson. A candidate improvement must be measured **in the estimand, the pipeline stage and the population where it will be deployed**. This one passed the floor, passed Bonferroni over seven arms, passed three seeds and passed a degeneracy check, and each of the three things that killed it was invisible to all of those: a pooling convention, a blend weight, and a calibration stage downstream that was already collecting the benefit. None of them is a statistical test.

7.11 The ceiling has a name: the light-order shortcut

The thirty-nine closed avenues above share a shape, and this subsection states it, because it is the most useful result in the report. The ceiling is not label noise and it is not the representation. It is a *shortcut shared by every fitted family*: predict the fencer whose lamp lit second.

Its size. The rule on its own scores 63.51% [61.5, 65.7] on the 2 631 pairs with a usable signed gap, and 61.18% [58.71, 63.70] as a two-feature model on 3 006 pairs. The production model predicts the second-lit side on **72.71**% of clips, so it does not merely use the rule, it **over-uses** it. The consequence splits the corpus in two:

subset	n	accuracy
the winner lit *second*	1 671	83.00%
the winner lit *first*	960	45.21%
gap		**+37.80 pp**

On the 960 the model is worse than a coin. That is where the ceiling lives.

It scales with the gap between the lamps. Binned by each clip's own frame rate, the gap grows monotonically: +26.59 *pp* below one frame, +38.16 at one to three, +44.45 above three. The long-minus-short contrast is **+17.39 pp** [+4.96, +29.41] with p(not increasing) = 0.003. Two qualifications belong with it. Only 2.62% of clips have $|dt| > 300$ ms, so the second lamp almost never falls outside the stored window; the mechanism is a *graduated* asymmetry, not a binary loss. And $|dt|$ also correlates with how separated, and therefore how legible, the action is, so the trend is *consistent with* window geometry without isolating it.

Where it lives, which decides what can fix it. The 157 temporal descriptors cut every window at the anchor, `seg = d[f_touch - w*fps : f_touch + 1]`, so they never see the segment after the first lamp; their share of the shortcut can only come from where the anchor sits. Measured separately:

model	accuracy	gap
frozen 256-column aggregate	67.16%	+39.65
157 temporal descriptors	64.58%	+18.03
full model	69.10%	+37.12

The aggregate block carries the shortcut nearly whole, and it is frozen: it is read from a cached artefact and is identical under every window variant. So a window fix can only reach the temporal block, which holds about half the effect.

Three window geometries, all null. Anchoring at the midpoint $f_{\text{first}} + |dt|/2$ makes the offset symmetric by construction. It moved the temporal gap by $-2.34\,pp$ $[-5.37, +0.96]$, crossing zero, at no cost in accuracy (+0.08) while a random-offset null did cost accuracy (−1.22) — a sanity check on the null, not support for the hypothesis. In the full model the midpoint made the gap slightly *worse*, +0.76, indistinguishable from its null at +0.99.

Anchoring *each fencer at its own lamp* is the only construction that equalises information content rather than merely offset, and it is available for 26 of the 157 columns: the 14 onset instants, whose helper already takes the anchor as an argument, and the 12 value-at-lamp columns. Of the rest, 48 signed extrema of a difference series are irreducibly joint (the frame of maximum $|d|$ cannot be formed from two separately windowed series), 48 window means are redefinable but become a different estimand, and 35 pair channels such as inter-fencer distance do not exist on a per-fencer grid. Result, paired, seeds 7/11/23, with the sign of each mechanism declared in advance:

model	arm	Δ mean	Δ gap
temporal	per-fencer, real	+0.30	+2.94
temporal	per-fencer, null	+0.27	+1.90
full	per-fencer, real	−0.11	−0.18
full	per-fencer, null	−0.11	+1.62

Excess of real over null on the mean: **+0.03 pp**. In the full model real and null are identical at −0.11. **Against the** 0.8 *pp* **floor there is no effect**, and this closure is firm rather than under-powered: the mean was resolvable at $\approx 1.3\,pp$ and the answer is no. The monotonicity route is a different matter and closes only for lack of power, with contrast intervals 4 to 10 points wide; there, "it did not move" remains indistinguishable from "I could not see it".

The reinterpretation, which is the actual finding. The 14 onset columns were not merely asymmetric, they were **mis-defined**: `d_t_adv` and its family mean "how long before its own lamp did this fencer start, extend, plant the front foot", and for the second-lit fencer they were measured against the *other* fencer's lamp. The pre-registered expectation was that fixing this would weaken the shortcut. It did the opposite: the gap *rose* by +2.94 *pp* with an interval excluding zero.

The explanation is uncomfortable and, we think, correct. A cleaner measurement of who moved first correlates *better* with who lit first. **So part of what this**

report has been calling a shortcut is the rule itself: in foil, the fencer who initiates does tend to light first and does tend to hold priority. The model was not purely cheating; it was using a legitimate correlate that happened to be badly anchored, and anchoring it correctly measures the same thing better. With its own null alongside at +1.90 the +2.94 does not survive as an effect (excess +1.04), so it is reported here as the interpretation that explains why no effect appears, not as an effect.

The arithmetic of the lever, and why it is not free. If the 960 were lifted to 60% the global figure would gain +5.40 *pp*; at 75%, +10.95 *pp*. Against that, dropping the 1 671 by five points costs −3.18 *pp* globally, so penalising the light order is a bad trade. The per-fencer anchor makes exactly the wrong exchange: it raises the 1 671 from 71.15 to 72.53% and lowers the 960 from 53.12 to 51.56%.

Why the 960 are a diagnostic and not a gate. The subset is defined as "the winner lit first", which requires knowing the winner, so it cannot be a routing condition at inference. It is also not a place to price a feature, and two independent measurements show why. In an optical-flow study an information-free block bought +3.08 *pp* [+1.83, +4.23] inside the region, and the real block's paired excess over it was +0.00 *pp*. In a video-transformer study a permuted block carrying zero information about the phrase bought **+0.47 pp** *inside* the region while *costing* −0.35 *pp* globally: points bought from nothing, precisely where the base sits at 45.19%. A subgroup selected by the models being wrong has a base below chance, and below chance any perturbation regresses upward. Throughout this subsection the verdict is the global paired mean and the split is description.

And the same trap in its other form. The complementary error is to condition on an action label that is itself defined by the outcome. For `accion = parada-respuesta` the modal scorer label is "simple attack with lunge", so the *scorer* attacked; for `accion = atacante` the loser's label is "attack parried". Conditioning fixes the correspondence between who extended and who won, and any feature measuring extension then predicts the label by construction: one flow feature reaches AUC = 0.895 inside blade actions, 0.224 outside and 0.360 unconditioned, the sign flipping with the label across all 14 venues. A spectacular 0.915 obtained that way was discarded. The 40 skeleton features score AUC = 0.067 on the 492 clips no family gets right, near-perfect inversion, because that set is *defined* by those models failing. None of these numbers describe a feature; they describe a selection.

The ceiling is not the representation, and that is measurable. It would be natural to conclude that pose is too impoverished and that pixels would carry the missing information. Measured, they do not, and the measurement is interpretable because its positive control passes: the same VideoMAE embeddings that add +0.08 *pp* [−0.51, +0.79] to priority predict the *venue* at 97.04% over 18 classes with a 14.47% majority, and the frame rate at 94.08%. The representation reads the video; it does not read the phrase.

Why the two differ is worth stating as a hypothesis, clearly marked as one. Venue is a global, static, high-variance property and survives averaging 1 568 tokens intact. Priority turns on a one-to-three-pixel object and a ten-to-thirty millisecond offset between extension and contact: small, local and fast, which is what an average destroys. Sixteen frames over 1.2 s is a 75 ms step, at or beyond the scale of the deciding event, and a 128 px corridor crop is below its spatial resolution.

Two measurements support the averaging account over a model-size account, and the first forecloses the obvious objection. **Capacity is not the limit.** Sweeping the retained components over two orders of magnitude gives a plateau, blade-action AUC 0.5674, 0.5486, 0.5686, **0.5924**, 0.5898, 0.5281 at $k = 8, 16, 32, 64, 128, 768$: the maximum sits at $k = 64$ and the full 768 *fall*, with the fraction of venues above chance collapsing to 0.820, the only point in the sweep that fails to separate. A ceiling reached already at eight components is a very-low-rank signal, not a signal awaiting more capacity. And the comparison that settles it: the best point the extra capacity buys is 0.5924, while the phase null at $k = 32$ — *all motion destroyed, static appearance kept* — already returns 0.5654. The 0.024 of AUC that capacity adds is the same size as what survives destroying the motion entirely, so **there is no k at which the physics appears**. In accuracy the best case over the full population is 63.72% against a 63.52% majority: two tenths.

Second, the averaged embedding is nearly invariant to circular reordering of its tokens, cosine 0.9980 against the real one, so it is reading level rather than temporal structure almost by construction. **The honest statement is not that the signal is absent from the pixels, but that VideoMAE-Base with a mean-pooled embedding does not capture it.** The indicated remedy would be attention instead of mean pooling, a narrow high-resolution box on the hands and blade, and dense sampling near the touch; on present evidence, which is 0.57 of AUC that a motion-destroying null matches, it is not recommended.

One precaution proved necessary and one leak proved absent. The interaction box contains the FIE scoreboard band in **2 272 of 3 006 clips** (75.6%), a median 18% of the crop, and at the touch anchor that band shows a green stripe on one side and a red stripe on the other: the lamps themselves, asymmetric by side, inside the input. They were masked. Re-running with the band visible gives +0.28 *pp* and AUC = 0.5631, no better than masked, so the shortcut was available in the pixels and this representation did not exploit it. Masking was obligatory anyway: an unexploited leak is not a safe leak.

What remains, and it is not a modelling change. The window ends `POST_S` = 0.3 s after the *first* lamp. That is short: in **104 of 3 330 clips** (3.12%) the second fencer's lamp falls outside the stored window entirely, so even the per-fencer anchor is clipped for those. The one geometric lever this report did not exhaust is therefore re-extraction with a longer `POST_S`, which is an acquisition change and not a modelling one. It is recorded as future work with that figure.

A decision recorded. Re-materialising the frozen aggregate over a new window was authorised in advance on one condition: a global mean delta clearing the 0.8 *pp* floor. The condition was not met, so the aggregates were left untouched, avoiding a re-materialisation and the invalidation of 23 091 cached artefacts. The value of the negative result is that the decision rests on a measurement of the one block that *could* be re-windowed, which moved nothing even when its columns were corrected.

8 Limitations

8.1 Statistical power is the binding constraint

This is the most important limitation in the report and it is not a modelling problem. With the held-out set as it stands, an exact McNemar test [7] on the conditional binomial with the observed discordance rate of 0.175 has 5.9% power to detect a true improvement of +1.6 *pp* — which happens to be the size of the last effect actually measured (Figure 38). Power reaches 80% only at about +8.5 *pp*; the minimum detectable effect at 80% power is 8.56 *pp*.

Real improvements of a point and a half are therefore *undetectable* with this corpus. That statement cuts both ways: it means most of the negative results in Section 7 are consistent with small real gains, and it means no positive result of that size can be believed either.

The remedy is not more models. Reaching 80% power for +1.6 *pp* needs about 5 478 test touches, against roughly 200 today — short by a factor of 15 to 25. Expressed in the unit that actually constrains the project, resampling whole tournaments gives 2.6% power with 2 tournaments, 54.4% with 12 and 80.3% with 20. At about 250 two-light pairs per major championship, that is **12 to 20 new major championships**. Everything else in this report is cheaper than that and none of it substitutes for it.

8.2 There is a ceiling, and adding models does not lift it

On the 3 047 development touches the tree and the temporal network are both right on 58.39%, exactly one is right on 25.47%, and *neither* is right on 16.15% (Figure 39). Inside the high-confidence band — blend probability above 0.80, $n = 1028$ — the both-wrong rate is still 12.2%.

Errors are correlated, which is why more models do not help: across ten model pairs, $P(A \text{ fails} \mid B \text{ fails})$ is 1.55 to 2.37 times what independence would give, and eight of the ten sit between 1.88 and 2.00. The oracle over the pool rises from 83.85% with two members to 94.03% with five while the best reachable blend stays at 73.88–73.94%; and four runs of the *same* network give an oracle of 83.95%, which is the cleanest demonstration that oracle inflation measures diversity of noise rather than recoverable signal.

8.3 No human reference exists

Nothing in this project has been validated against expert human judgement. The labels are the archive's record of who was awarded the touch. The action categories are checked only for agreement between two code paths that read the same poses. The κ values of Section 6.1 are reproducibility, and the gold directory in the codebase contains scripts and no annotations. Any claim about refereeing accuracy is therefore out of scope, and an independent human annotation on a separate sample is the single change that would most improve the evidential status of this work.

8.4 Generalisation and acquisition remain confounded

Restating Section 2.3 as a limitation: because both held-out tournaments are blind venues, the held-out accuracy cannot separate a failure to generalise from a worse-placed touch instant. Protocol C attributes about 1.4 of the 4.93 *pp* gap to the anchor and leaves the rest unattributed. Fixing the five overlay geometries would resolve this, and it is the cheapest high-value item on the list.

8.5 An identified defect that is not fixed

The on-guard segment that supplies the on-guard geometry of Section 3.4 is anchored to index 0 of the stored array rather than to the touch. This is the case in six modules — `armwork.py:39`, `calibration.py:66`, `clip_stats.py:28`, `distance_control.py:83`, `footwork.py:64` and `piste_viz.py:50` — of which only `calibration.py:312` adapts the offset when the anchor moves.

For clips stored as a window this is nearly harmless: index 0 is about four seconds before the touch and still inside the phrase. For clips stored at full duration it is not, because the on-guard position is then read from the beginning of the recording, which may be a different phrase entirely.

This is recorded as a defect and not as a result, for two reasons. The quantities that would describe its magnitude were produced outside the reports this document cites, so they are not quoted (Section 8.6). And the patch that would correct it **is not applied**: it was written, applied and then reverted, because it fails the production test battery with two failures and fourteen

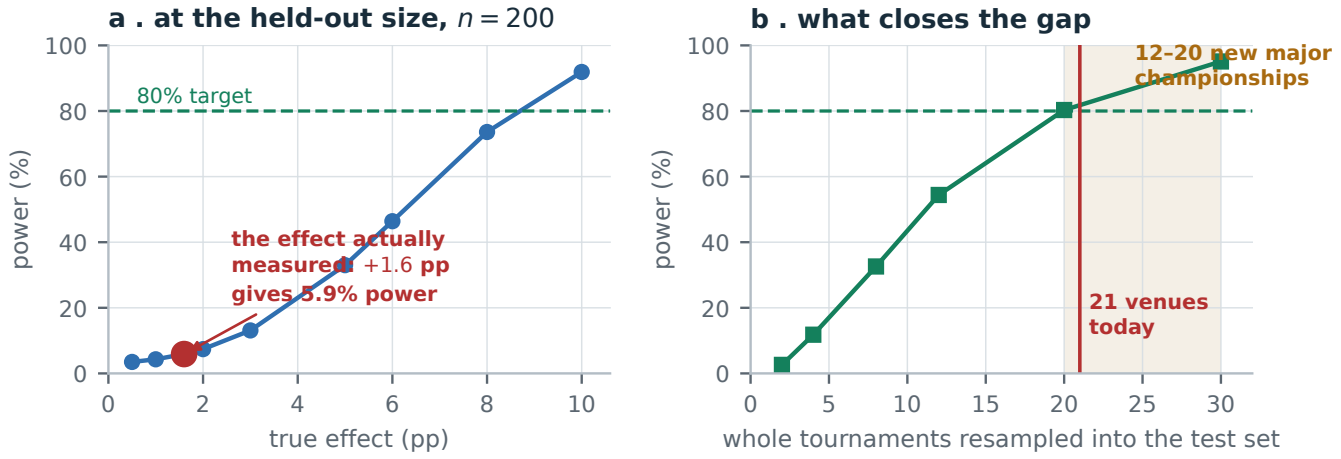


Figure 38. Statistical power. **(a)** Power against true effect at the held-out size, by exact McNemar on the conditional binomial with the observed discordance rate. **(b)** Power against the number of whole tournaments resampled into the test set. The corpus is short by a factor of 15 to 25 in touches, or 12 to 20 major championships.

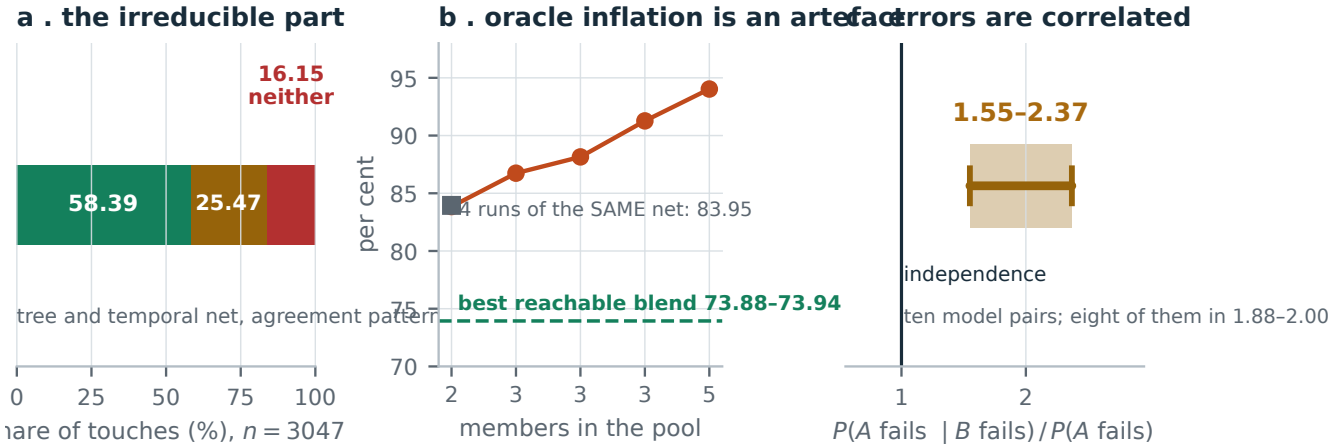


Figure 39. The ceiling. **(a)** Agreement pattern of the two production models on 3 047 touches. **(b)** Oracle accuracy grows with the pool while the best reachable blend does not, and four runs of the same network inflate the oracle almost as much as four different families. **(c)** Failure correlation between model pairs, relative to independence.

errors, having introduced a module that is neither declared in the producer fingerprint list of Section 2 nor shipped in the portable package. Every on-guard quantity in this report — the guard-distance distribution of Figure 16 and the guard-line separation of Section 6.4 — is therefore computed with the defect present, and should be read that way.

8.6 Numbers this report does not use

Several figures circulating in earlier internal notes could not be traced to a measurement report during the preparation of this document, and they have been left out rather than repeated. They are listed so their absence is a decision:

- A forearm-angle error of 38.67° median and 152° maximum attributed to the on-guard anchoring defect, together with a 546/546 backward-compatibility check over 40 clips and a causality check on the single clip with a zero anchor. The defect itself is confirmed and is reported in Section 8.5; these particular quantities were produced outside the reports cited here, so the results section that would have contained them was cut rather than sourced to a document a reader cannot inspect. The re-extraction identity checks that *are* in the record are cited in Section 3.5 instead.
- A keypoint displacement of 0.125 in normalised coordinates attributed to changing the pose batch size. Not found, and the value occurs only twice in the whole record, both times as a Cohen κ in unrelated tables. What is measured is determinism at batch 16, with no comparison against any other batch value, as Section 3.5 now states explicitly.
- A stature coverage of "404 of 423 fencers, with 30 of 63 488 appearances uncovered". This is a real measurement, but of a third quantity: the *availability* of a published or inferred height per name, rather than the method a clip used. The corpus contains 401 fencers and all 401 resolve, so the availability figure is not the one that bears on whether the scale is contaminated; Section 6.3 reports the two that are.
- A -6.7% scale bias attributed to the sex fallback.

The mechanism is documented but is recorded as not firing on the clips in question, so no bias of that size is claimed here. What Section 6.3 reports instead is the per-side and per-clip exposure, both of which are measured.

- A single global figure of "the real accuracy". There is no protocol under which such a number exists, which is the reason for Section 5.4.

9 Conclusion and future work

A retrospective classifier of foil priority built from broadcast video reaches, on 18 development tournaments with one tournament held out at a time and no abstention, 73.3–73.65% as a method and 73.88% as one frozen artefact, against 69.68% for the decision tree it replaces: an improvement of about $+3.7\,pp$, roughly 4.6 times the measured noise floor. Allowed to abstain under a conformal gate, the same system is right on 80.5% of the two-light touches it answers at 51.4% coverage, and cuts confidently wrong verdicts from 12.10% to 2.80% — with the caveat, which is not a footnote, that the guarantee is marginal and per-venue accuracy when speaking ranges from 64.8% to 91.2%. Measured across all 21 tournaments, the five venues whose scoreboard overlay cannot be read cost $4.93\,pp$ $[2.14, 6.97]$ of accuracy in those venues.

The negative results are the larger contribution. Thirty-nine avenues were built and closed against explicit null controls, and the recurring pattern is worth stating on its own: in a corpus this size, a null construction gains $+0.69\,pp$, a random per-clip duration is worth as much as the correct one, shuffled transfer labels are worth as much as real ones, and a real venue-specific gain can be $+1.32\,pp$ where it is not needed and $+0.00\,pp$ where it is tested. Without a measured floor and a matched control for every avenue, most of these would have been reported as improvements.

Three things would change the picture, in order of expected value per unit of effort. First, **five table entries** for the four overlay geometries of Section 2.2, which is a bounded engineering task that recovers a measured $4.93\,pp$ at five venues and unconfounds the held-out split. Second, an **independent human annotation** on a separate sample, which is the only route to any claim about refereeing rather than reproducibility. Third, and dominating both in the long run, **12 to 20 new major championships**, because at 21 tournaments the power to detect a $+1.6\,pp$ improvement is 5.9% and no amount of modelling changes that number.

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Table 5. The ablation ledger. "Effect" is the measured change against the stated reference; "control" is the same construction run on data that cannot carry the information. An avenue is closed when the effect fails to clear the 0.8 *pp* floor of Section 4.4 or fails to beat its control. References differ between rows, so effects are not comparable across rows as accuracies. ‡ marks a row whose original control the audit of Section 4.5 found defective for that block's feature type, and whose control column now reports the corrected side-sign null instead. Note that a side-sign null *harms* (−1.03) where the floor-calibrating null *helps* (+0.69), so the excess of real over control on these rows is not comparable with the 0.8 *pp* floor; the floor is applied to the effect column, as everywhere else in this table.

Avenue	Protocol and n	Effect	Control	Why it closed
Audio: date the blade clash	100 clips, 11 venues	—	—	recovers a guaranteed event on 20–28% at a 10% false-alarm point
Audio: stereo panorama	77 of 100 clips	—	—	$\rho = +0.002$ ($p = 0.829$) where the control recovers 0.303
Adaptive window	LOTO, 3 130 pairs	+0.19	+0.00	a random duration from the same distribution scores the same
Window trimmed to 2.3 s	500 re-extracted clips	—	0.0000	236 of 413 features break; 91.9% silent collapse
Full-duration context	1 022 clips, 15 venues	−2.45	+1.08	null blocks beat the real ones
Macro window summaries	LOTO, 3 130 pairs	+0.22	—	inside the floor
dt_{ms} as a model input	LOTO, 3 047 pairs	−0.68	−0.65	identical to its shuffled twin
dt_{ms} as auxiliary target	LOTO, 3 047 pairs	+1.15	−0.49	gain is +1.32 at readable venues, +0.00 at generic ones
Simultaneity as gate signal	12 600 clips	—	—	model confidence beats it by 14 points; ships disabled
Rulebook features, 51 cols	GroupKFold, 3 059 pairs	+0.66	+0.76	does not beat its own shifted null
Cross-body features, 52 cols	LOTO, 3 203 clips	+0.16	−1.09	inside the floor; $r = 0.979$ with existing columns
wrist-gap sub-family	LOTO, 3 203 clips	−0.86	−0.64	worse than its own nulls
Piste-position features, 30 cols	LOTO, 3 130 pairs	−0.26	+0.06	below its null and below its noised self
Laterality pairing, 4 cols	LOTO, 3 130 pairs	−0.35	−0.40	symmetric under the mirror by construction
Transfer from one-light clips	11 460 + 3 130 clips	+1.06	+1.06	shuffled labels give the identical gain
No-point clips	16 933 index records	—	—	the class does not exist in the label field
Correction clips	34 374 manifest rows	—	—	the archive contradicts itself on who scored
Competition level	LOTO, 3 118 pairs	−1.97	—	effect triples if the class boundary moves
Referee-criterion drift	LOTO, 3 130 pairs	−0.35	−0.34	placebo season performs identically
A third model family (best of 4)	LOTO, 3 047 pairs	+0.08	—	each needs 3–5 *pp* more to earn any weight
Stacking instead of a fixed weight	LOTO, 3 047 pairs	−0.72	—	every combination form is below the fixed weight
Blend weight 0.65 → 0.85	LOTO, worst venue	+3.76	−4.24	sign flips once the gate is recalibrated
Extra runs in the blend	LOTO, 3 047 pairs	+0.45	—	inside the floor, at +33% inference each
Network capacity ×4.5	LOTO, 3 047 pairs	+0.03	—	inside the floor
Invariance augmentation	4 transforms, 18 venues	+0.92	—	does not reach two standard errors
Multitask action taxonomy	LOTO, 3 047 pairs	+0.30	+0.40	shuffled labels score higher
Monocular depth for scale	425 frames, 100 clips	—	—	46–51% error on a known 4.00 m span
Piste line detection	500 frames, 19 venues	3.8%	4.3%	already at the assisted-human ceiling
Deciding the 871 two-label clips	194 clips with usable dt	—	—	$p = 0.112$ and 0.725; 9 and 10 in the minority class
Data augmentation	2 clips, 20 transforms	—	—	attempted, no accuracy consequence measured
Added 9 September 2026. Paired design: same folds, same seeds 7/11/23, only the block changes.				
Front-foot landing, rule t.83, 10 cols	LOTO, 3 041 pairs	−1.12	−0.76	worse than both its nulls; landing is resolved for both fencers in 81.7% of clips, so coverage is not the limit, signal is
Centre of mass, *temps d'arrêt*, 24 cols	LOTO, 3 041 pairs	+0.39	−1.03‡	**closed by redundancy, not by its null**: the block beats a side-destroyed copy of itself by one to two points, so the signal is real, but it is already inside via $r = 0.877$, 0.841 and 0.802 with existing columns, and the block's own delta stays inside the floor
pull sub-family, 6 cols	LOTO, 3 041 pairs	−0.92	−0.49‡	below the floor, and below its healthy null too (excess −0.43)
All three blocks together, 40 cols	LOTO, 3 041 pairs	−0.03	−0.41‡	the combination does not rescue the parts; excess over its healthy null is +0.37, inside the floor
Antisymmetric siamese latent	smoke test, 2 mirrors	—	—	the penalty falls 88 → 30 with